Distinctive Image Features from Scale-Invariant Keypoints

David G. Lowe
Computer Science Department
University of British Columbia
Vancouver, B.C., Canada
lowe@cs.ubc.ca

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Abstract

This paper presents a method for extracting distinctive invariant features from images that can be used to perform reliable matching between different views of an object or scene. The features are invariant to image scale and rotation, and are shown to provide robust matching across a a substantial range of affine distortion, change in 3D viewpoint, addition of noise, and change in illumination. The features are highly distinctive, in the sense that a single feature can be correctly matched with high probability against a large database of features from many images. This paper also describes an approach to using these features for object recognition. The recognition proceeds by matching individual features to a database of features from known objects using a fast nearest-neighbor algorithm, followed by a Hough transform to identify clusters belonging to a single object, and finally performing verification through least-squares solution for consistent pose parameters. This approach to recognition can robustly identify objects among clutter and occlusion while achieving near real-time performance.

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尺度不变关键点中的独特图像特征

David G. Lowe　计算机
科学系　不列颠哥伦比亚大学
温哥华，B.C.，加拿大lowe
@cs.ubc.ca

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摘要

本文提出了一种从图像中提取独特不变特征的方法，该特征可用于在不同视角的对象或场景之间进行可靠匹配。这些特征对图像尺度和旋转具有不变性，并能在较大范围的仿射畸变、三维视角变化、噪声干扰以及光照变化下实现稳健匹配。这些特征具有高度独特性，单个特征能够以高概率正确匹配来自大量图像的特征数据库。本文还描述了利用这些特征进行物体识别的方法。识别过程首先通过快速最近邻算法将单个特征与已知物体的特征数据库进行匹配，随后采用霍夫变换识别属于同一物体的特征簇，最后通过最小二乘解算一致姿态参数进行验证。该识别方法能够在杂乱和遮挡环境中稳健地识别物体，并实现近实时性能。

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1 Introduction

Image matching is a fundamental aspect of many problems in computer vision, including object or scene recognition, solving for 3D structure from multiple images, stereo correspondence, and motion tracking. This paper describes image features that have many properties that make them suitable for matching differing images of an object or scene. The features are invariant to image scaling and rotation, and partially invariant to change in illumination and 3D camera viewpoint. They are well localized in both the spatial and frequency domains, reducing the probability of disruption by occlusion, clutter, or noise. Large numbers of features can be extracted from typical images with efficient algorithms. In addition, the features are highly distinctive, which allows a single feature to be correctly matched with high probability against a large database of features, providing a basis for object and scene recognition.

The cost of extracting these features is minimized by taking a cascade filtering approach, in which the more expensive operations are applied only at locations that pass an initial test. Following are the major stages of computation used to generate the set of image features:

1. **Scale-space extrema detection:** The first stage of computation searches over all scales and image locations. It is implemented efficiently by using a difference-of-Gaussian function to identify potential interest points that are invariant to scale and orientation.

2. **Keypoint localization:** At each candidate location, a detailed model is fit to determine location and scale. Keypoints are selected based on measures of their stability.

3. **Orientation assignment:** One or more orientations are assigned to each keypoint location based on local image gradient directions. All future operations are performed on image data that has been transformed relative to the assigned orientation, scale, and location for each feature, thereby providing invariance to these transformations.

4. **Keypoint descriptor:** The local image gradients are measured at the selected scale in the region around each keypoint. These are transformed into a representation that allows for significant levels of local shape distortion and change in illumination.

This approach has been named the Scale Invariant Feature Transform (SIFT), as it transforms image data into scale-invariant coordinates relative to local features.

An important aspect of this approach is that it generates large numbers of features that densely cover the image over the full range of scales and locations. A typical image of size 500x500 pixels will give rise to about 2000 stable features (although this number depends on both image content and choices for various parameters). The quantity of features is particularly important for object recognition, where the ability to detect small objects in cluttered backgrounds requires that at least 3 features be correctly matched from each object for reliable identification.

For image matching and recognition, SIFT features are first extracted from a set of reference images and stored in a database. A new image is matched by individually comparing each feature from the new image to this previous database and finding candidate matching features based on Euclidean distance of their feature vectors. This paper will discuss fast nearest-neighbor algorithms that can perform this computation rapidly against large databases.

The keypoint descriptors are highly distinctive, which allows a single feature to find its correct match with good probability in a large database of features. However, in a cluttered

1 引言

图像匹配是计算机视觉中许多问题的基本方面，包括物体或场景识别、从多幅图像求解三维结构、立体对应以及运动跟踪。本文描述的图像特征具有多种特性，使其适用于匹配同一物体或场景的不同图像。这些特征对图像缩放和旋转具有不变性，对照明变化和三维摄像机视角变化具有部分不变性。它们在空间域和频域中都能被精确定位，降低了因遮挡、杂乱或噪声干扰的可能性。通过高效算法可从典型图像中提取大量特征。此外，这些特征具有高度独特性，使得单个特征能够以高概率正确匹配大型特征数据库，从而为物体和场景识别提供基础。

提取这些特征的成本通过采用级联过滤方法得以最小化，该方法中更昂贵的操作仅应用于通过初始测试的位置。以下是用于生成图像特征集的主要计算阶段：

1. 尺度空间极值检测：计算的第一阶段在所有尺度和图像位置进行搜索。通过使用高斯差分函数高效实现，以识别对尺度和方向不变的特征点。

2. 关键点定位：在每个候选位置，拟合一个详细模型以确定位置和尺度。关键点的选择基于其稳定性的度量。

3. 方向分配：基于局部图像梯度方向，为每个关键点位置分配一个或多个方向。所有后续操作均在相对于每个特征所分配的方向、尺度和位置进行变换后的图像数据上执行，从而确保对这些变换的不变性。

4. 关键点描述符：在选定尺度下，测量每个关键点周围区域的局部图像梯度。这些梯度被转换为一种能够显著容忍局部形状畸变和光照变化的表示形式。

该方法被命名为尺度不变特征变换（SIFT），因为它将图像数据转换为相对于局部特征的尺度不变坐标。

这种方法的一个重要方面是它能生成大量特征，这些特征在图像的所有尺度和位置上密集覆盖。一张典型的500x500像素图像会产生约2000个稳定特征（尽管具体数量取决于图像内容及各项参数的选择）。特征数量对于物体识别尤为重要——在杂乱背景中检测小物体时，每个物体至少需要正确匹配3个特征才能实现可靠识别。

对于图像匹配与识别，首先从一组参考图像中提取SIFT特征并存储于数据库中。新图像的匹配通过将其每个特征与此前数据库逐一比对，并基于特征向量的欧几里得距离寻找候选匹配特征来实现。本文将探讨能够针对大型数据库快速执行此计算的最近邻算法。

关键点描述符具有高度独特性，这使得单个特征能够在大型特征数据库中，以较高的概率找到其正确匹配。然而，在杂乱环境中

image, many features from the background will not have any correct match in the database, giving rise to many false matches in addition to the correct ones. The correct matches can be filtered from the full set of matches by identifying subsets of keypoints that agree on the object and its location, scale, and orientation in the new image. The probability that several features will agree on these parameters by chance is much lower than the probability that any individual feature match will be in error. The determination of these consistent clusters can be performed rapidly by using an efficient hash table implementation of the generalized Hough transform.

Each cluster of 3 or more features that agree on an object and its pose is then subject to further detailed verification. First, a least-squared estimate is made for an affine approximation to the object pose. Any other image features consistent with this pose are identified, and outliers are discarded. Finally, a detailed computation is made of the probability that a particular set of features indicates the presence of an object, given the accuracy of fit and number of probable false matches. Object matches that pass all these tests can be identified as correct with high confidence.

2 Related research

The development of image matching by using a set of local interest points can be traced back to the work of Moravec (1981) on stereo matching using a corner detector. The Moravec detector was improved by Harris and Stephens (1988) to make it more repeatable under small image variations and near edges. Harris also showed its value for efficient motion tracking and 3D structure from motion recovery (Harris, 1992), and the Harris corner detector has since been widely used for many other image matching tasks. While these feature detectors are usually called corner detectors, they are not selecting just corners, but rather any image location that has large gradients in all directions at a predetermined scale.

The initial applications were to stereo and short-range motion tracking, but the approach was later extended to more difficult problems. Zhang *et al.* (1995) showed that it was possible to match Harris corners over a large image range by using a correlation window around each corner to select likely matches. Outliers were then removed by solving for a fundamental matrix describing the geometric constraints between the two views of rigid scene and removing matches that did not agree with the majority solution. At the same time, a similar approach was developed by Torr (1995) for long-range motion matching, in which geometric constraints were used to remove outliers for rigid objects moving within an image.

The ground-breaking work of Schmid and Mohr (1997) showed that invariant local feature matching could be extended to general image recognition problems in which a feature was matched against a large database of images. They also used Harris corners to select interest points, but rather than matching with a correlation window, they used a rotationally invariant descriptor of the local image region. This allowed features to be matched under arbitrary orientation change between the two images. Furthermore, they demonstrated that multiple feature matches could accomplish general recognition under occlusion and clutter by identifying consistent clusters of matched features.

The Harris corner detector is very sensitive to changes in image scale, so it does not provide a good basis for matching images of different sizes. Earlier work by the author (Lowe, 1999) extended the local feature approach to achieve scale invariance. This work also described a new local descriptor that provided more distinctive features while being less

图像中，背景的许多特征在数据库中将没有任何正确匹配，除了正确的匹配外还会产生许多错误匹配。通过识别在新图像中关于物体及其位置、尺度和方向一致的关键点子集，可以从完整匹配集中筛选出正确匹配。多个特征在这些参数上偶然一致的概率远低于任何单个特征匹配出错的概率。通过使用广义霍夫变换的高效哈希表实现，可以快速确定这些一致的聚类。

对于每个包含3个或以上特征、且在物体及其姿态上达成一致的簇，会进行进一步的详细验证。首先，通过最小二乘法估算物体姿态的仿射近似值。接着，识别出与此姿态一致的其他图像特征，并剔除异常值。最后，根据拟合精度和可能误匹配的数量，详细计算特定特征集指示物体存在的概率。通过所有这些测试的物体匹配结果，可以高置信度地判定为正确。

2 相关研究

利用一组局部兴趣点进行图像匹配的发展可以追溯到Moravec（1981）使用角点检测器进行立体匹配的工作。Harris和Stephens（1988）改进了Moravec检测器，使其在微小图像变化和边缘附近更具可重复性。Harris还展示了其在高效运动跟踪和运动恢复三维结构中的价值（Harris，1992），此后哈里斯角点检测器被广泛用于许多其他图像匹配任务。虽然这些特征检测器通常被称为角点检测器，但它们选择的不仅仅是角点，而是在预定尺度上所有方向都具有大梯度的任何图像位置。

最初的应用是针对立体视觉和短距离运动追踪，但该方法后来被扩展到更复杂的问题上。Zhang *et al.* (1995) 证明，通过在每个角点周围使用相关窗口来选择可能的匹配，可以在大范围的图像中匹配哈里斯角点。然后，通过求解描述刚性场景两个视图之间几何约束的基本矩阵，并移除与多数解不一致的匹配点，从而剔除异常值。与此同时，Torr（1995）为长距离运动匹配开发了类似的方法，其中利用几何约束来移除图像内刚性物体运动产生的异常值。

Schmid和Mohr（1997）的开创性工作表明，不变局部特征匹配可以扩展到一般图像识别问题中，即在一个大型图像数据库中匹配特征。他们同样采用Harris角点来选取兴趣点，但并未使用相关窗口进行匹配，而是采用了局部图像区域的旋转不变描述符。这使得特征能够在两幅图像之间发生任意方向变化时仍可匹配。此外，他们证明通过识别匹配特征的一致性聚类，多重特征匹配能够在遮挡和杂波干扰下实现通用识别。

Harris角点检测器对图像尺度的变化非常敏感，因此无法为匹配不同尺寸的图像提供良好基础。作者早期的工作（Lowe，1999）扩展了局部特征方法以实现尺度不变性。该研究还描述了一种新的局部描述符，能在保持较低{v*}的同时提供更具区分性的特征。

sensitive to local image distortions such as 3D viewpoint change. This current paper provides a more in-depth development and analysis of this earlier work, while also presenting a number of improvements in stability and feature invariance.

There is a considerable body of previous research on identifying representations that are stable under scale change. Some of the first work in this area was by Crowley and Parker (1984), who developed a representation that identified peaks and ridges in scale space and linked these into a tree structure. The tree structure could then be matched between images with arbitrary scale change. More recent work on graph-based matching by Shokoufandeh, Marsic and Dickinson (1999) provides more distinctive feature descriptors using wavelet coefficients. The problem of identifying an appropriate and consistent scale for feature detection has been studied in depth by Lindeberg (1993, 1994). He describes this as a problem of scale selection, and we make use of his results below.

Recently, there has been an impressive body of work on extending local features to be invariant to full affine transformations (Baumberg, 2000; Tuytelaars and Van Gool, 2000; Mikolajczyk and Schmid, 2002; Schaffalitzky and Zisserman, 2002; Brown and Lowe, 2002). This allows for invariant matching to features on a planar surface under changes in orthographic 3D projection, in most cases by resampling the image in a local affine frame. However, none of these approaches are yet fully affine invariant, as they start with initial feature scales and locations selected in a non-affine-invariant manner due to the prohibitive cost of exploring the full affine space. The affine frames are are also more sensitive to noise than those of the scale-invariant features, so in practice the affine features have lower repeatability than the scale-invariant features unless the affine distortion is greater than about a 40 degree tilt of a planar surface (Mikolajczyk, 2002). Wider affine invariance may not be important for many applications, as training views are best taken at least every 30 degrees rotation in viewpoint (meaning that recognition is within 15 degrees of the closest training view) in order to capture non-planar changes and occlusion effects for 3D objects.

While the method to be presented in this paper is not fully affine invariant, a different approach is used in which the local descriptor allows relative feature positions to shift significantly with only small changes in the descriptor. This approach not only allows the descriptors to be reliably matched across a considerable range of affine distortion, but it also makes the features more robust against changes in 3D viewpoint for non-planar surfaces. Other advantages include much more efficient feature extraction and the ability to identify larger numbers of features. On the other hand, affine invariance is a valuable property for matching planar surfaces under very large view changes, and further research should be performed on the best ways to combine this with non-planar 3D viewpoint invariance in an efficient and stable manner.

Many other feature types have been proposed for use in recognition, some of which could be used in addition to the features described in this paper to provide further matches under differing circumstances. One class of features are those that make use of image contours or region boundaries, which should make them less likely to be disrupted by cluttered backgrounds near object boundaries. Matas *et al.,* (2002) have shown that their maximally-stable extremal regions can produce large numbers of matching features with good stability. Mikolajczyk *et al.,* (2003) have developed a new descriptor that uses local edges while ignoring unrelated nearby edges, providing the ability to find stable features even near the boundaries of narrow shapes superimposed on background clutter. Nelson and Selinger (1998) have shown good results with local features based on groupings of image contours. Similarly,

对局部图像失真（如3D视角变化）敏感。本文在先前研究基础上进行了更深入的拓展与分析，同时提出了多项在稳定性和特征不变性方面的改进。

关于识别在尺度变化下保持稳定的表示方法，已有大量先前研究。该领域的早期工作之一是Crowley和Parker（1984）提出的，他们开发了一种能识别尺度空间中的峰值与脊线、并将其连接成树状结构的表示方法。这种树结构可在任意尺度变化的图像间进行匹配。Shokoufandeh、Marsic和Dickinson（1999）近期基于图匹配的研究，利用小波系数提供了更具区分度的特征描述符。Lindeberg（1993, 1994）则深入研究了为特征检测确定合适且一致尺度的问题，他将此称为尺度选择问题，下文我们将运用其研究成果。

近年来，在将局部特征扩展至完全仿射变换不变性方面涌现了大量令人瞩目的研究成果（Baumberg, 2000; Tuytelaars and Van Gool, 2000; Mikolajczyk and Schmid, 2002 ; Schaffalitzky and Zisserman, 2002; Brown and Lowe, 2002）。这类方法通过局部仿射框架对图像进行重采样，在多数情况下能够实现平面表面特征在正交三维投影变化下的不变匹配。然而，由于完整探索仿射空间的计算成本过高，这些方法均以非仿射不变的方式选择初始特征尺度与位置，因而尚未实现完全的仿射不变性。相较于尺度不变特征，仿射框架对噪声更为敏感，因此实践中除非平面倾斜角度超过约40度（Mikolajczyk, 2002），否则仿射特征的重复性往往低于尺度不变特征。对于许多应用场景而言，更广泛的仿射不变性可能并非必需——为捕捉三维物体的非平面形变与遮挡效应，最佳训练视角通常需至少每隔30度采集一次（这意味着识别视角与最近训练视角的偏差需控制在15度以内）。

尽管本文提出的方法并非完全仿射不变，但我们采用了一种不同的策略，使得局部描述子允许特征相对位置在描述子仅发生微小变化时产生显著偏移。这种方法不仅能让描述子在较大范围的仿射畸变下实现可靠匹配，还能使特征对非平面表面的三维视角变化更具鲁棒性。其他优势包括更高效的特征提取能力以及识别更多特征的可能性。另一方面，仿射不变性对于在极大视角变化下匹配平面表面具有重要价值，未来研究应进一步探索如何以高效稳定的方式，将这种特性与非平面的三维视角不变性相结合。

许多其他特征类型已被提议用于识别，其中一些可以结合本文所述的特征使用，以在不同情况下提供更多匹配。一类特征是利用图像轮廓或区域边界的特征，这应使它们不太可能被物体边界附近杂乱的背景干扰。Matas *et al.,* (2002) 已证明，他们的最大稳定极值区域能够产生大量具有良好稳定性的匹配特征。Mikolajczyk *et al.,* (2003) 开发了一种新的描述符，该描述符利用局部边缘同时忽略无关的邻近边缘，从而即使在叠加于杂乱背景上的狭窄形状边界附近也能找到稳定特征。Nelson 和 Selinger（1998）展示了基于图像轮廓分组的局部特征的良好效果。类似地，

Pope and Lowe (2000) used features based on the hierarchical grouping of image contours, which are particularly useful for objects lacking detailed texture.

The history of research on visual recognition contains work on a diverse set of other image properties that can be used as feature measurements. Carneiro and Jepson (2002) describe phase-based local features that represent the phase rather than the magnitude of local spatial frequencies, which is likely to provide improved invariance to illumination. Schiele and Crowley (2000) have proposed the use of multidimensional histograms summarizing the distribution of measurements within image regions. This type of feature may be particularly useful for recognition of textured objects with deformable shapes. Basri and Jacobs (1997) have demonstrated the value of extracting local region boundaries for recognition. Other useful properties to incorporate include color, motion, figure-ground discrimination, region shape descriptors, and stereo depth cues. The local feature approach can easily incorporate novel feature types because extra features contribute to robustness when they provide correct matches, but otherwise do little harm other than their cost of computation. Therefore, future systems are likely to combine many feature types.

3 Detection of scale-space extrema

As described in the introduction, we will detect keypoints using a cascade filtering approach that uses efficient algorithms to identify candidate locations that are then examined in further detail. The first stage of keypoint detection is to identify locations and scales that can be repeatably assigned under differing views of the same object. Detecting locations that are invariant to scale change of the image can be accomplished by searching for stable features across all possible scales, using a continuous function of scale known as scale space (Witkin, 1983).

It has been shown by Koenderink (1984) and Lindeberg (1994) that under a variety of reasonable assumptions the only possible scale-space kernel is the Gaussian function. Therefore, the scale space of an image is defined as a function, $L(x, y, \sigma)$, that is produced from the convolution of a variable-scale Gaussian, $G(x, y, \sigma)$, with an input image, $I(x, y)$:

$$L(x, y, \sigma) = G(x, y, \sigma) * I(x, y),$$

where $*$ is the convolution operation in x and y, and

$$G(x, y, \sigma) = \frac{1}{2\pi\sigma^2} e^{-(x^2+y^2)/2\sigma^2}.$$

To efficiently detect stable keypoint locations in scale space, we have proposed (Lowe, 1999) using scale-space extrema in the difference-of-Gaussian function convolved with the image, $D(x, y, \sigma)$, which can be computed from the difference of two nearby scales separated by a constant multiplicative factor k:

$$\begin{aligned} D(x, y, \sigma) &= (G(x, y, k\sigma) - G(x, y, \sigma)) * I(x, y) \\ &= L(x, y, k\sigma) - L(x, y, \sigma). \end{aligned} \quad (1)$$

There are a number of reasons for choosing this function. First, it is a particularly efficient function to compute, as the smoothed images, L, need to be computed in any case for scale space feature description, and D can therefore be computed by simple image subtraction.

Pope和Lowe（2000）采用了基于图像轮廓层次化分组的特征，这些特征对于缺乏细节纹理的物体尤为有用。

视觉识别研究的历史包含了多种其他图像属性的研究工作，这些属性可作为特征度量使用。Carneiro和Jepson（2002）描述了基于相位的局部特征，这类特征表示局部空间频率的相位而非幅度，可能提供更好的光照不变性。Schiele和Crowley（2000）提出了使用多维直方图来概括图像区域内测量值的分布。此类特征可能对识别具有可变形形状的纹理物体特别有用。Basri和Jacobs（1997）论证了提取局部区域边界对识别的价值。其他值得整合的有用属性包括颜色、运动、图-底区分、区域形状描述符以及立体深度线索。局部特征方法能轻松整合新型特征类型，因为当额外特征提供正确匹配时会增强鲁棒性，即使匹配失败，除了计算成本外几乎不会造成负面影响。因此，未来的系统很可能将融合多种特征类型。

3 尺度空间极值检测

如引言所述，我们将采用级联过滤方法检测关键点，该方法通过高效算法识别候选位置，随后进行更详细的检测。关键点检测的第一阶段是确定在不同视角下能够被稳定重复定位的位置与尺度。要检测对图像尺度变化具有不变性的位置，可通过尺度空间（Witkin, 1983）这一连续尺度函数，在所有可能的尺度中搜索稳定的特征来实现。

Koenderink（1984）和Lindeberg（1994）的研究表明，在多种合理的假设下，唯一可能的尺度空间核是高斯函数。因此，图像的尺度空间被定义为一个函数$L(x, y, \sigma)$，它是由可变尺度的高斯函数$G(x, y, \sigma)$与输入图像$I(x, y)$进行卷积产生的：

$$L(x, y, \sigma) = G(x, y, \sigma) * I(x, y),$$

其中 $*$ 是 x 和 y 中的卷积运算，且

$$G(x, y, \sigma) = \frac{1}{2\pi\sigma^2} e^{-(x^2+y^2)/2\sigma^2}.$$

为了在尺度空间中高效检测稳定的关键点位置，我们曾提出（Lowe, 1999）利用图像与高斯差分函数卷积后的尺度空间极值$D(x, y, \sigma)$，该函数可通过两个相邻尺度（其间以恒定乘积因子k分隔）的差值计算得出：

$$\begin{aligned} D(x, y, \sigma) &= (G(x, y, k\sigma) - G(x, y, \sigma)) * I(x, y) \\ &= L(x, y, k\sigma) - L(x, y, \sigma). \end{aligned} \quad (1)$$

选择此函数的原因有很多。首先，它是一个计算效率特别高的函数，因为平滑图像 L, 无论如何都需要为尺度空间特征描述而计算，因此 D 可以通过简单的图像相减来计算。

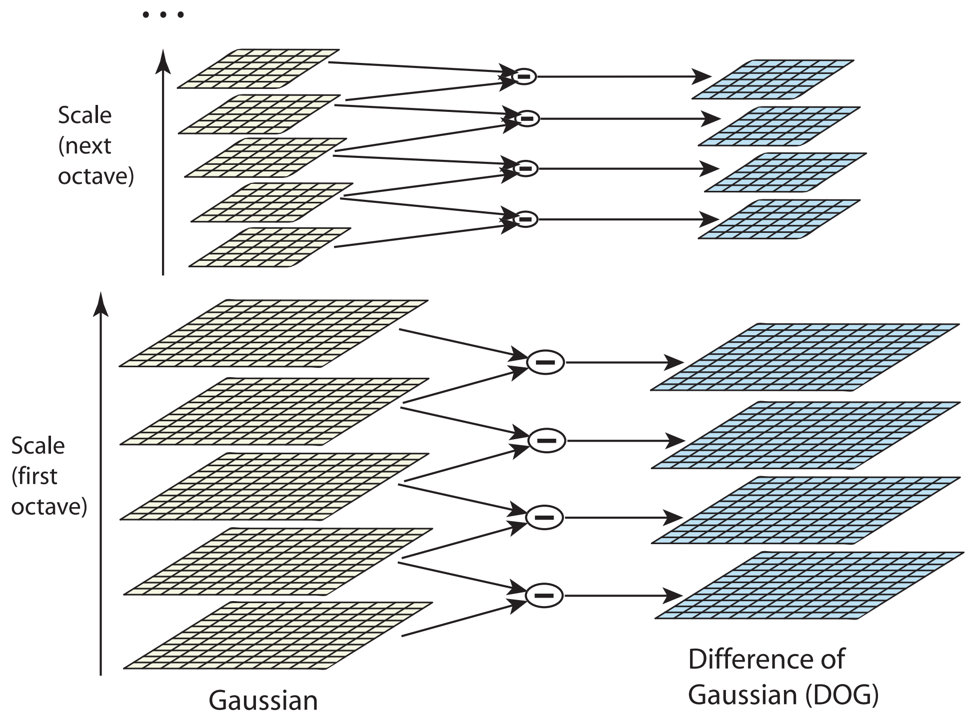

Figure 1: For each octave of scale space, the initial image is repeatedly convolved with Gaussians to produce the set of scale space images shown on the left. Adjacent Gaussian images are subtracted to produce the difference-of-Gaussian images on the right. After each octave, the Gaussian image is down-sampled by a factor of 2, and the process repeated.

In addition, the difference-of-Gaussian function provides a close approximation to the scale-normalized Laplacian of Gaussian, $\sigma^2\nabla^2 G$, as studied by Lindeberg (1994). Lindeberg showed that the normalization of the Laplacian with the factor σ^2 is required for true scale invariance. In detailed experimental comparisons, Mikolajczyk (2002) found that the maxima and minima of $\sigma^2\nabla^2 G$ produce the most stable image features compared to a range of other possible image functions, such as the gradient, Hessian, or Harris corner function.

The relationship between D and $\sigma^2\nabla^2 G$ can be understood from the heat diffusion equation (parameterized in terms of σ rather than the more usual $t = \sigma^2$):

$$\frac{\partial G}{\partial \sigma} = \sigma \nabla^2 G.$$

From this, we see that $\nabla^2 G$ can be computed from the finite difference approximation to $\partial G/\partial\sigma$, using the difference of nearby scales at $k\sigma$ and σ:

$$\sigma\nabla^2 G = \frac{\partial G}{\partial \sigma} \approx \frac{G(x, y, k\sigma) - G(x, y, \sigma)}{k\sigma - \sigma}$$

and therefore,

$$G(x, y, k\sigma) - G(x, y, \sigma) \approx (k-1)\sigma^2\nabla^2 G.$$

This shows that when the difference-of-Gaussian function has scales differing by a constant factor it already incorporates the σ^2 scale normalization required for the scale-invariant

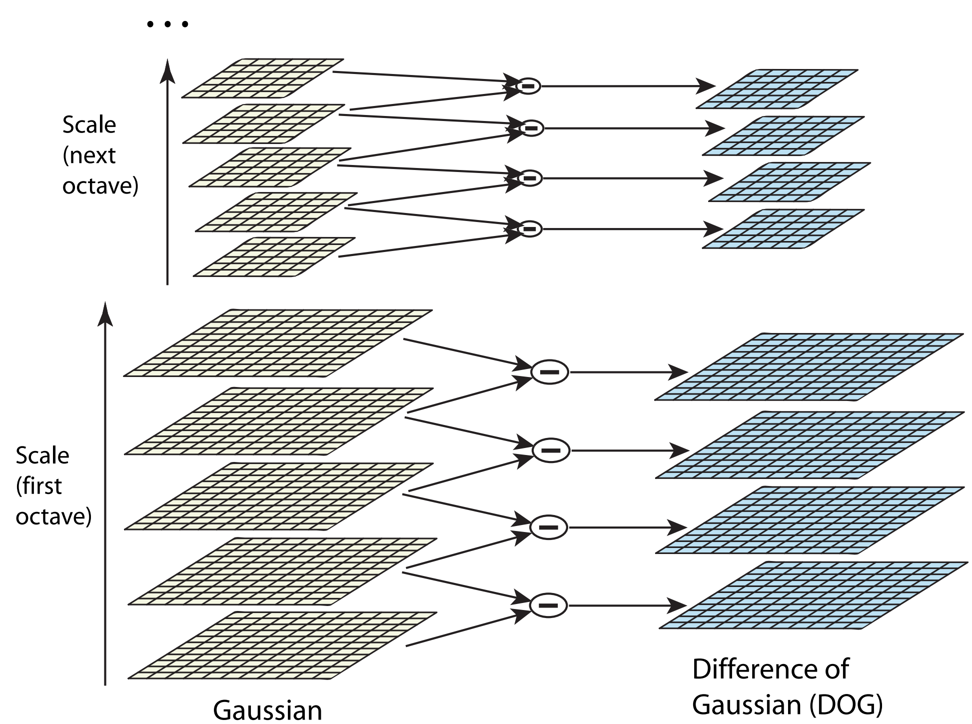

图1：对于尺度空间的每个八度，初始图像与高斯函数反复卷积，生成左侧所示的尺度空间图像集。相邻的高斯图像相减后，生成右侧的高斯差分图像。每个八度结束后，高斯图像会进行2倍降采样，并重复此过程。

此外，差分高斯函数为尺度归一化的拉普拉斯高斯函数{v*}提供了紧密的近似，正如Lindeberg（1994）所研究的那样。Lindeberg证明，拉普拉斯算子需通过因子{v*}进行归一化才能实现真正的尺度不变性。在详细的实验比较中，Mikolajczyk（2002）发现，相较于梯度、海森矩阵或哈里斯角点函数等一系列其他可能的图像函数，{v*}的极大值和极小值能产生最稳定的图像特征。

D与$\sigma^2\nabla^2 G$之间的关系可通过热扩散方程（以σ而非更常见的$t=\sigma^2$为参数）来理解：

$$\frac{\partial G}{\partial \sigma} = \sigma \nabla^2 G.$$

由此，我们看到 $\nabla^2 G$ 可以通过对 $\partial G/\partial\sigma$ 的有限差分近似来计算，利用 $k\sigma$ 和 σ 处邻近尺度的差值：

$$\sigma \nabla^2 G = \frac{\partial G}{\partial \sigma} \approx \frac{G(x,y,k\sigma) - G(x,y,\sigma)}{k\sigma - \sigma}$$

因此，

$$G(x,y,k\sigma) - G(x,y,\sigma) \approx (k-1)\sigma^2 \nabla^2 G.$$

这表明，当高斯差分函数的尺度相差一个常数因子时，它已经包含了尺度不变性所需的 σ^2 尺度归一化。

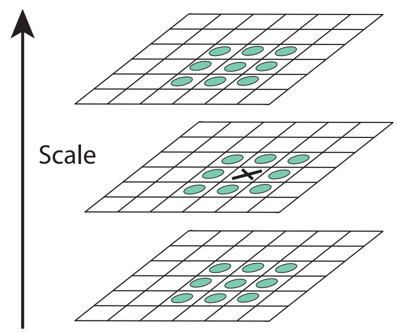

Figure 2: Maxima and minima of the difference-of-Gaussian images are detected by comparing a pixel (marked with X) to its 26 neighbors in 3x3 regions at the current and adjacent scales (marked with circles).

Laplacian. The factor $(k-1)$ in the equation is a constant over all scales and therefore does not influence extrema location. The approximation error will go to zero as k goes to 1, but in practice we have found that the approximation has almost no impact on the stability of extrema detection or localization for even significant differences in scale, such as $k = \sqrt{2}$.

An efficient approach to construction of $D(x, y, \sigma)$ is shown in Figure 1. The initial image is incrementally convolved with Gaussians to produce images separated by a constant factor k in scale space, shown stacked in the left column. We choose to divide each octave of scale space (i.e., doubling of σ) into an integer number, s, of intervals, so $k = 2^{1/s}$. We must produce $s + 3$ images in the stack of blurred images for each octave, so that final extrema detection covers a complete octave. Adjacent image scales are subtracted to produce the difference-of-Gaussian images shown on the right. Once a complete octave has been processed, we resample the Gaussian image that has twice the initial value of σ (it will be 2 images from the top of the stack) by taking every second pixel in each row and column. The accuracy of sampling relative to σ is no different than for the start of the previous octave, while computation is greatly reduced.

3.1 Local extrema detection

In order to detect the local maxima and minima of $D(x, y, \sigma)$, each sample point is compared to its eight neighbors in the current image and nine neighbors in the scale above and below (see Figure 2). It is selected only if it is larger than all of these neighbors or smaller than all of them. The cost of this check is reasonably low due to the fact that most sample points will be eliminated following the first few checks.

An important issue is to determine the frequency of sampling in the image and scale domains that is needed to reliably detect the extrema. Unfortunately, it turns out that there is no minimum spacing of samples that will detect all extrema, as the extrema can be arbitrarily close together. This can be seen by considering a white circle on a black background, which will have a single scale space maximum where the circular positive central region of the difference-of-Gaussian function matches the size and location of the circle. For a very elongated ellipse, there will be two maxima near each end of the ellipse. As the locations of maxima are a continuous function of the image, for some ellipse with intermediate elongation there will be a transition from a single maximum to two, with the maxima arbitrarily close to

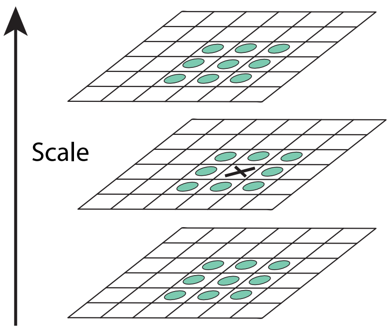

图2：高斯差分图像的最大值和最小值是通过将像素（标记为X）与其在当前尺度及相邻尺度下3x3区域中的26个邻域像素（标记为圆圈）进行比较来检测的。

拉普拉斯算子。方程中的因子$(k-1)$在所有尺度上都是常数，因此不影响极值位置。当k趋近于1时，近似误差将趋于零，但实际上我们发现即使尺度差异显著（例如$k=\sqrt{2}$），这种近似对极值检测或定位的稳定性也几乎没有影响。

一种构建$D(x,y,\sigma)$的高效方法如图1所示。初始图像与高斯函数进行增量卷积，以在尺度空间中生成间隔为恒定因子k的图像，如左列堆叠所示。我们选择将尺度空间的每个八度（即σ的翻倍）划分为整数个s区间，因此$k\ = 2^{1/s}$.。我们必须为每个八度生成$s+3$幅模糊图像堆栈，以确保最终的极值检测覆盖完整的八度。相邻尺度的图像相减后，得到右侧所示的高斯差分图像。当一个完整八度处理完毕后，我们对高斯图像进行重采样——该图像的σ (初始值将加倍（即位于堆栈顶部向下第2幅图像））——方法是在每行和每列中隔一个像素取样。相对于σ的采样精度与前一八度起始时相同，而计算量则大幅减少。

3.1 局部极值检测

为了检测$D(x,y,\sigma)$的局部极大值和极小值，每个采样点需与当前图像中的八个相邻点以及上下相邻尺度中的九个相邻点进行比较（见图2）。仅当该点大于所有这些相邻点或小于所有相邻点时，才会被选中。由于大多数采样点会在最初几次比较后被淘汰，这一检测过程的计算成本相对较低。

一个重要的问题是确定在图像和尺度域中采样的频率，以便可靠地检测极值点。不幸的是，事实证明，不存在能够检测所有极值点的最小采样间隔，因为极值点可以任意接近。这可以通过考虑黑色背景上的白色圆形来理解：当高斯差分函数的圆形正中心区域与圆的大小和位置匹配时，尺度空间中将存在一个最大值。对于一个非常细长的椭圆，其两端附近将出现两个最大值。由于最大值的位置是图像的连续函数，对于某些中等细长程度的椭圆，将会出现从一个最大值到两个最大值的过渡，且这两个最大值可以任意接近。

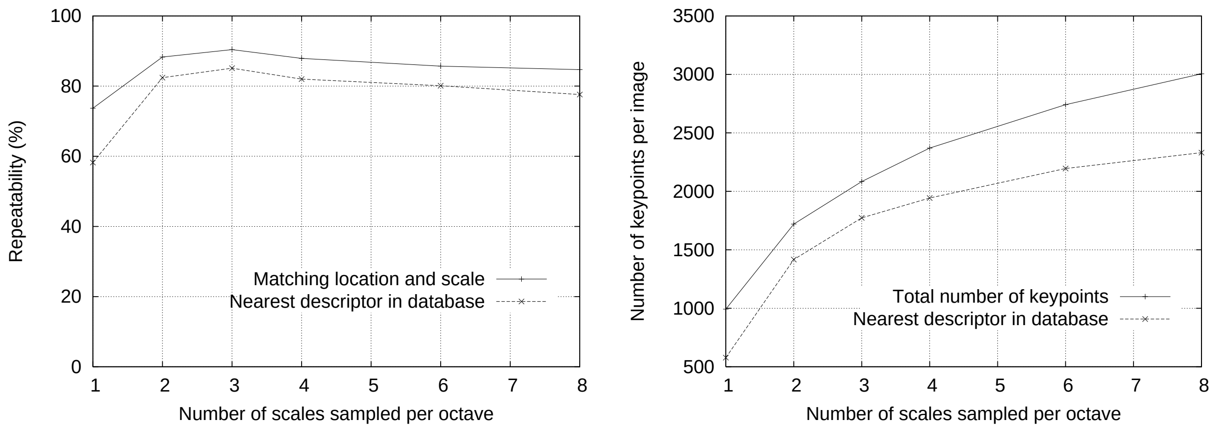

Figure 3: The top line of the first graph shows the percent of keypoints that are repeatably detected at the same location and scale in a transformed image as a function of the number of scales sampled per octave. The lower line shows the percent of keypoints that have their descriptors correctly matched to a large database. The second graph shows the total number of keypoints detected in a typical image as a function of the number of scale samples.

each other near the transition.

Therefore, we must settle for a solution that trades off efficiency with completeness. In fact, as might be expected and is confirmed by our experiments, extrema that are close together are quite unstable to small perturbations of the image. We can determine the best choices experimentally by studying a range of sampling frequencies and using those that provide the most reliable results under a realistic simulation of the matching task.

3.2 Frequency of sampling in scale

The experimental determination of sampling frequency that maximizes extrema stability is shown in Figures 3 and 4. These figures (and most other simulations in this paper) are based on a matching task using a collection of 32 real images drawn from a diverse range, including outdoor scenes, human faces, aerial photographs, and industrial images (the image domain was found to have almost no influence on any of the results). Each image was then subject to a range of transformations, including rotation, scaling, affine stretch, change in brightness and contrast, and addition of image noise. Because the changes were synthetic, it was possible to precisely predict where each feature in an original image should appear in the transformed image, allowing for measurement of correct repeatability and positional accuracy for each feature.

Figure 3 shows these simulation results used to examine the effect of varying the number of scales per octave at which the image function is sampled prior to extrema detection. In this case, each image was resampled following rotation by a random angle and scaling by a random amount between 0.2 of 0.9 times the original size. Keypoints from the reduced resolution image were matched against those from the original image so that the scales for all keypoints would be be present in the matched image. In addition, 1% image noise was added, meaning that each pixel had a random number added from the uniform interval [-0.01,0.01] where pixel values are in the range [0,1] (equivalent to providing slightly less than 6 bits of accuracy for image pixels).

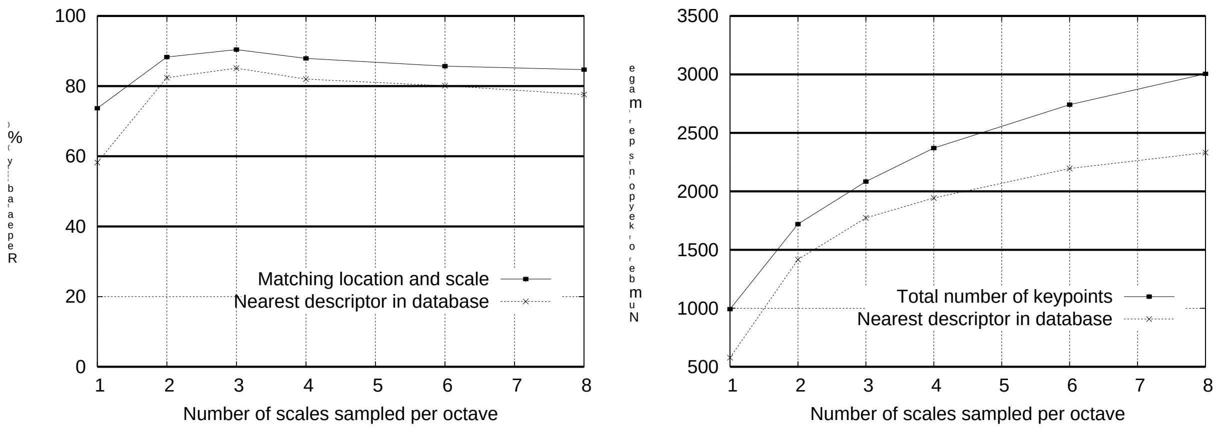

图3：第一张图表的顶线显示了在变换图像中，在相同位置和尺度上可重复检测到的关键点百分比，作为每八度音阶采样尺度数量的函数。底线显示了那些描述符能够与大型数据库正确匹配的关键点百分比。第二张图表展示了典型图像中检测到的关键点总数，作为尺度采样数量的函数。

在转变附近彼此接近。

因此，我们必须接受一种在效率与完整性之间权衡的解决方案。事实上，正如预期且经实验证实的那样，彼此接近的极值点对图像的微小扰动非常不稳定。我们可以通过研究一系列采样频率，并在匹配任务的实际模拟中使用那些能提供最可靠结果的频率，从而通过实验确定最佳选择。

3.2 尺度中的采样频率

最大化极值稳定性的采样频率实验测定结果如图3和图4所示。这些图表（及本文大多数其他仿真实验）基于一项匹配任务，该任务使用了32幅真实图像组成的多样化集合，涵盖户外场景、人脸、航拍图像和工业图像（研究发现图像域对结果几乎不产生影响）。每幅图像随后经过一系列变换处理，包括旋转、缩放、仿射拉伸、亮度与对比度调整以及图像噪声添加。由于这些变化是合成生成的，我们能够精确预测原始图像中每个特征在变换后图像中的应现位置，从而实现对每个特征正确重复率与位置精度的量化测量。

图3展示了这些模拟结果，用于研究在极值检测前对图像函数进行采样时，每八度音阶尺度数量变化的影响。在此情况下，每张图像先经过随机角度旋转和随机比例缩放（缩放比例在原始尺寸的0.2至0.9倍之间），随后进行重采样。将降分辨率图像中的关键点与原始图像中的关键点进行匹配，以确保所有关键点的尺度都能在匹配图像中体现。此外，还添加了1%的图像噪声，这意味着每个像素值在[0,1]范围内（相当于为图像像素提供略低于6位的精度）的基础上，叠加了来自均匀区间[-0.01,0.01]的随机数。

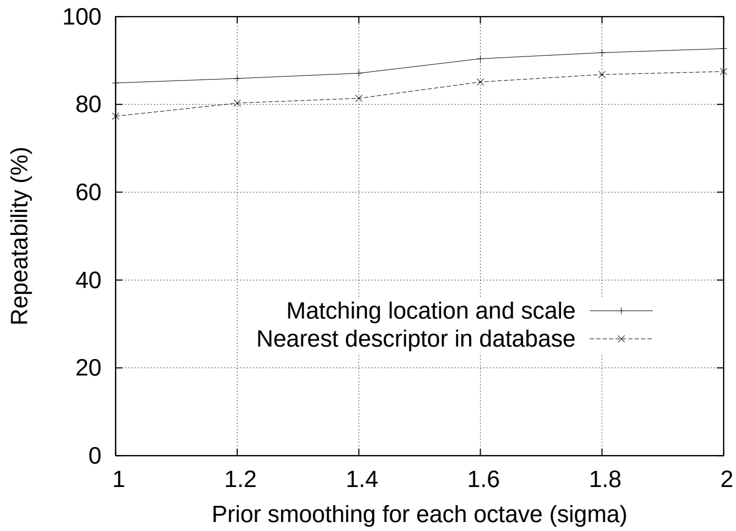

Figure 4: The top line in the graph shows the percent of keypoint locations that are repeatably detected in a transformed image as a function of the prior image smoothing for the first level of each octave. The lower line shows the percent of descriptors correctly matched against a large database.

The top line in the first graph of Figure 3 shows the percent of keypoints that are detected at a matching location and scale in the transformed image. For all examples in this paper, we define a matching scale as being within a factor of $\sqrt{2}$ of the correct scale, and a matching location as being within σ pixels, where σ is the scale of the keypoint (defined from equation (1) as the standard deviation of the smallest Gaussian used in the difference-of-Gaussian function). The lower line on this graph shows the number of keypoints that are correctly matched to a database of 40,000 keypoints using the nearest-neighbor matching procedure to be described in Section 6 (this shows that once the keypoint is repeatably located, it is likely to be useful for recognition and matching tasks). As this graph shows, the highest repeatability is obtained when sampling 3 scales per octave, and this is the number of scale samples used for all other experiments throughout this paper.

It might seem surprising that the repeatability does not continue to improve as more scales are sampled. The reason is that this results in many more local extrema being detected, but these extrema are on average less stable and therefore are less likely to be detected in the transformed image. This is shown by the second graph in Figure 3, which shows the average number of keypoints detected and correctly matched in each image. The number of keypoints rises with increased sampling of scales and the total number of correct matches also rises. Since the success of object recognition often depends more on the quantity of correctly matched keypoints, as opposed to their percentage correct matching, for many applications it will be optimal to use a larger number of scale samples. However, the cost of computation also rises with this number, so for the experiments in this paper we have chosen to use just 3 scale samples per octave.

To summarize, these experiments show that the scale-space difference-of-Gaussian function has a large number of extrema and that it would be very expensive to detect them all. Fortunately, we can detect the most stable and useful subset even with a coarse sampling of scales.

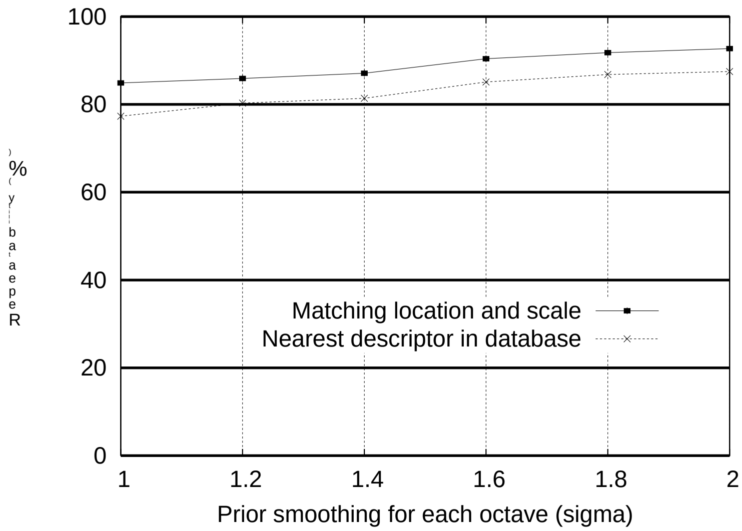

图4：图中顶线显示了在变换图像中可重复检测到的关键点位置百分比，作为每个八度第一层先验图像平滑度的函数。底线显示了与大型数据库正确匹配的描述符百分比。

图3第一张图中的顶线显示了在变换图像中，在匹配位置和尺度上检测到的关键点百分比。对于本文中的所有示例，我们将匹配尺度定义为在正确尺度的$\sqrt{2}$倍以内，匹配位置定义为在σ像素以内，其中σ是关键点的尺度（根据公式(1)定义为高斯差分函数中使用的最小高斯的标准差）。该图中的底线显示了使用第6节将描述的最近邻匹配程序，与包含40,000个关键点的数据库正确匹配的关键点数量（这表明一旦关键点可重复定位，它很可能对识别和匹配任务有用）。如图所示，当每个八度采样3个尺度时，可获得最高的重复性，这也是本文所有其他实验所使用的尺度采样数量。

随着采样尺度的增加，可重复性并未持续提升，这或许令人惊讶。原因在于，这会导致检测到更多的局部极值点，但这些极值点的平均稳定性较低，因此在变换后的图像中被检测到的可能性较小。图3中的第二张图表展示了这一点，该图显示了每张图像中检测到并正确匹配的关键点的平均数量。随着尺度采样的增加，关键点数量上升，正确匹配的总数也随之增加。由于物体识别的成功往往更依赖于正确匹配关键点的数量，而非其正确匹配的百分比，因此在许多应用中，使用更多尺度采样样本将是最优选择。然而，计算成本也会随此数量增加，因此本文实验中，我们选择每八度仅使用3个尺度采样样本。

总而言之，这些实验表明，尺度空间的高斯差分函数{v*}存在大量极值点，检测所有极值点的计算成本极高。幸运的是，即使对尺度进行粗采样，我们也能检测到其中最稳定且最有用的子集。

3.3 Frequency of sampling in the spatial domain

Just as we determined the frequency of sampling per octave of scale space, so we must determine the frequency of sampling in the image domain relative to the scale of smoothing. Given that extrema can be arbitrarily close together, there will be a similar trade-off between sampling frequency and rate of detection. Figure 4 shows an experimental determination of the amount of prior smoothing, σ, that is applied to each image level before building the scale space representation for an octave. Again, the top line is the repeatability of keypoint detection, and the results show that the repeatability continues to increase with σ. However, there is a cost to using a large σ in terms of efficiency, so we have chosen to use $\sigma = 1.6$, which provides close to optimal repeatability. This value is used throughout this paper and was used for the results in Figure 3.

Of course, if we pre-smooth the image before extrema detection, we are effectively discarding the highest spatial frequencies. Therefore, to make full use of the input, the image can be expanded to create more sample points than were present in the original. We double the size of the input image using linear interpolation prior to building the first level of the pyramid. While the equivalent operation could effectively have been performed by using sets of subpixel-offset filters on the original image, the image doubling leads to a more efficient implementation. We assume that the original image has a blur of at least $\sigma = 0.5$ (the minimum needed to prevent significant aliasing), and that therefore the doubled image has $\sigma = 1.0$ relative to its new pixel spacing. This means that little additional smoothing is needed prior to creation of the first octave of scale space. The image doubling increases the number of stable keypoints by almost a factor of 4, but no significant further improvements were found with a larger expansion factor.

4 Accurate keypoint localization

Once a keypoint candidate has been found by comparing a pixel to its neighbors, the next step is to perform a detailed fit to the nearby data for location, scale, and ratio of principal curvatures. This information allows points to be rejected that have low contrast (and are therefore sensitive to noise) or are poorly localized along an edge.

The initial implementation of this approach (Lowe, 1999) simply located keypoints at the location and scale of the central sample point. However, recently Brown has developed a method (Brown and Lowe, 2002) for fitting a 3D quadratic function to the local sample points to determine the interpolated location of the maximum, and his experiments showed that this provides a substantial improvement to matching and stability. His approach uses the Taylor expansion (up to the quadratic terms) of the scale-space function, $D(x, y, \sigma)$, shifted so that the origin is at the sample point:

$$D(\mathbf{x}) = D + \frac{\partial D}{\partial \mathbf{x}}^T \mathbf{x} + \frac{1}{2}\mathbf{x}^{\mathbf{T}} \frac{\partial^2 D}{\partial \mathbf{x}^2}\mathbf{x} \tag{2}$$

where D and its derivatives are evaluated at the sample point and $\mathbf{x} = (x, y, \sigma)^T$ is the offset from this point. The location of the extremum, $\hat{\mathbf{x}}$, is determined by taking the derivative of this function with respect to $\mathbf{x}$ and setting it to zero, giving

$$\hat{\mathbf{x}} = -\frac{\partial^2 D}{\partial \mathbf{x}^2}^{-1} \frac{\partial D}{\partial \mathbf{x}}. \tag{3}$$

3.3 空间域采样频率

正如我们确定了尺度空间每八度的采样频率一样，我们也必须确定图像域中相对于平滑尺度的采样频率。鉴于极值点可以任意接近，采样频率与检测率之间将存在类似的权衡。图4展示了在构建八度尺度空间表示之前，对每个图像层级应用的先验平滑量σ的实验测定结果。同样，顶线表示关键点检测的重复性，结果显示重复性随σ的增加而持续提升。然而，使用较大的σ会带来效率上的代价，因此我们选择使用σ =1.6，这提供了接近最优的重复性。该值在本文中全程使用，并已应用于图3的结果。

当然，如果在极值检测前对图像进行预平滑处理，我们实际上就丢弃了最高的空间频率。因此，为了充分利用输入信息，可以通过扩展图像来生成比原始图像更多的采样点。我们在构建金字塔第一层之前，使用线性插值将输入图像的尺寸扩大一倍。虽然通过在原始图像上使用亚像素偏移滤波器组也能实现等效操作，但图像加倍能带来更高效的实现。我们假设原始图像至少具有σ =0.5的模糊度（这是避免明显混叠所需的最小值），因此加倍后的图像相对于其新像素间距具有σ =1.0的模糊度。这意味着在创建尺度空间的第一阶八度前，几乎不需要额外的平滑处理。图像加倍使稳定关键点的数量增加近4倍，但采用更大的扩展系数并未带来显著改善。

4 精确关键点定位

一旦通过比较像素与其邻域找到关键点候选，下一步便是对附近数据进行详细拟合，以确定位置、尺度以及主曲率之比。这些信息可用于剔除低对比度（因而对噪声敏感）或边缘定位不佳的点。

该方法的最初实现（Lowe，1999）仅将关键点定位在中心采样点的位置和尺度上。然而，Brown最近开发了一种方法（Brown and Lowe，2002），通过对局部采样点拟合三维二次函数来确定最大值的插值位置，其实验表明这显著提升了匹配性能和稳定性。他的方法利用了尺度空间函数{v*}的泰勒展开（至二次项），并将原点平移至采样点:

$$D(\mathbf{x}) = D + \frac{\partial D}{\partial \mathbf{x}}^T \mathbf{x} + \frac{1}{2}\mathbf{x}^\mathbf{T} \frac{\partial^2 D}{\partial \mathbf{x}^2} \mathbf{x} \tag{2}$$

其中 D 及其导数在采样点处取值，$\mathbf{x} = (x, y, \sigma)^T$ 是该点的偏移量。极值点 $\hat{\mathbf{x}}$ 的位置通过对该函数关于 $\mathbf{x}$ 求导并令其为零来确定，从而得到

$$\hat{\mathbf{x}} = -\frac{\partial^2 D}{\partial \mathbf{x}^2}^{-1} \frac{\partial D}{\partial \mathbf{x}}. \tag{3}$$

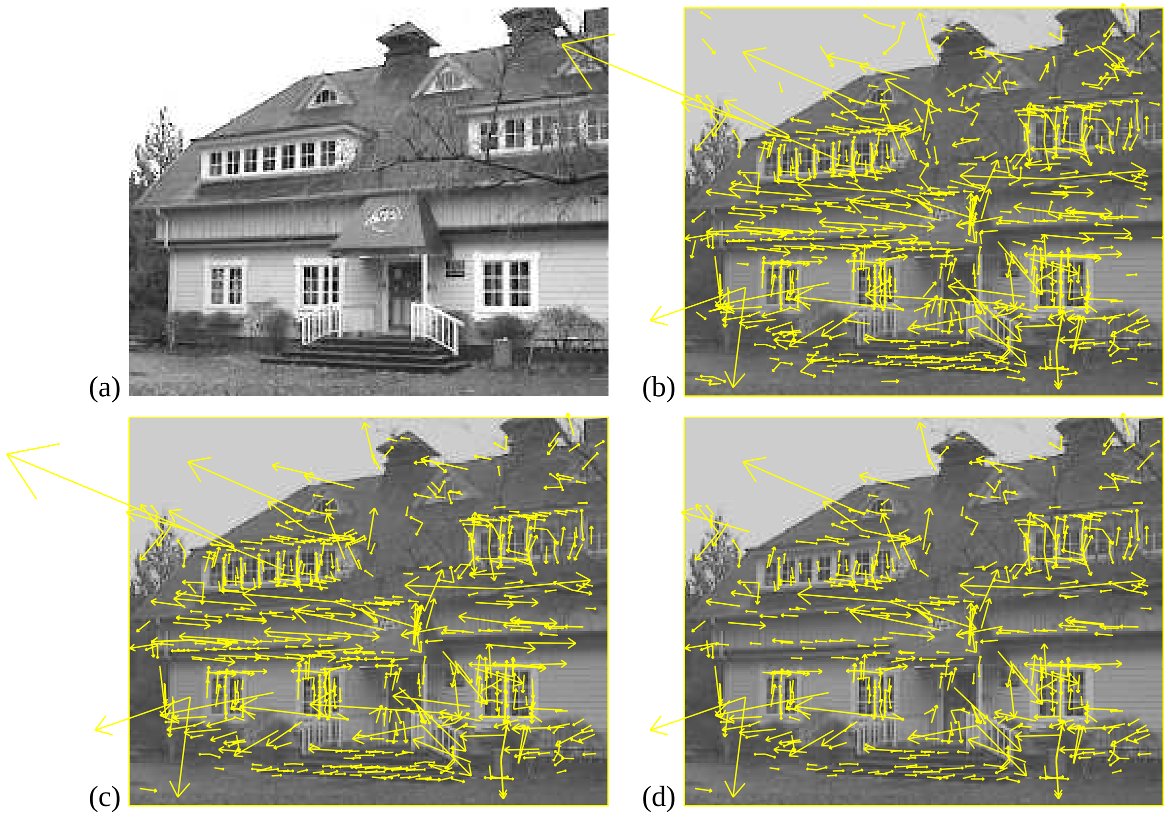

Figure 5: This figure shows the stages of keypoint selection. (a) The 233x189 pixel original image. (b) The initial 832 keypoints locations at maxima and minima of the difference-of-Gaussian function. Keypoints are displayed as vectors indicating scale, orientation, and location. (c) After applying a threshold on minimum contrast, 729 keypoints remain. (d) The final 536 keypoints that remain following an additional threshold on ratio of principal curvatures.

As suggested by Brown, the Hessian and derivative of D are approximated by using differences of neighboring sample points. The resulting 3x3 linear system can be solved with minimal cost. If the offset $\hat{\mathbf{x}}$ is larger than 0.5 in any dimension, then it means that the extremum lies closer to a different sample point. In this case, the sample point is changed and the interpolation performed instead about that point. The final offset $\hat{\mathbf{x}}$ is added to the location of its sample point to get the interpolated estimate for the location of the extremum.

The function value at the extremum, $D(\hat{\mathbf{x}})$, is useful for rejecting unstable extrema with low contrast. This can be obtained by substituting equation (3) into (2), giving

$$D(\hat{\mathbf{x}}) = D + \frac{1}{2}\frac{\partial D}{\partial \mathbf{x}}^T \hat{\mathbf{x}}.$$

For the experiments in this paper, all extrema with a value of $|D(\hat{\mathbf{x}})|$ less than 0.03 were discarded (as before, we assume image pixel values in the range [0,1]).

Figure 5 shows the effects of keypoint selection on a natural image. In order to avoid too much clutter, a low-resolution 233 by 189 pixel image is used and keypoints are shown as vectors giving the location, scale, and orientation of each keypoint (orientation assignment is described below). Figure 5 (a) shows the original image, which is shown at reduced contrast behind the subsequent figures. Figure 5 (b) shows the 832 keypoints at all detected maxima

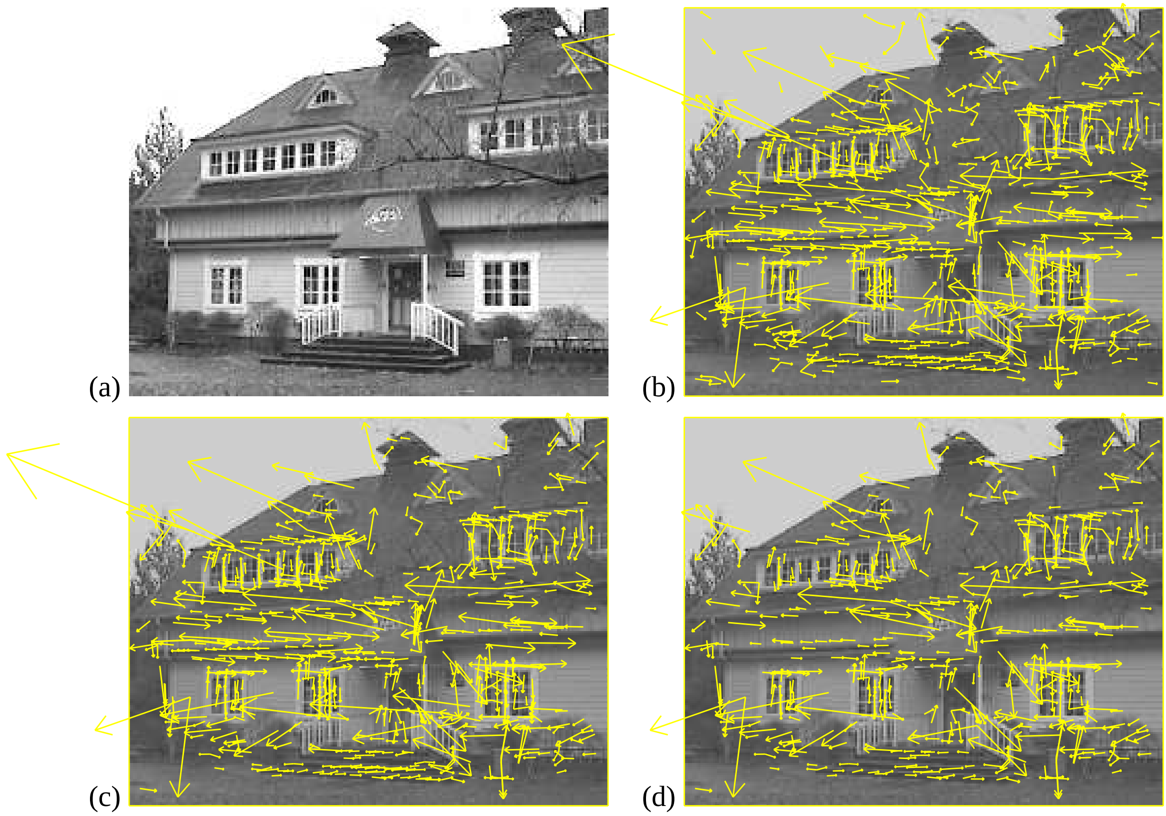

图5：此图展示了关键点选择的各个阶段。(a) 233x189像素的原始图像。(b) 高斯差分函数极值处的832个初始关键点位置，关键点以表示尺度、方向和位置的矢量形式显示。(c) 应用最小对比度阈值后，剩余729个关键点。(d) 经过主曲率比值附加阈值处理后，最终保留的536个关键点。

正如Brown所建议，通过使用相邻采样点之间的差异来近似D的Hessian矩阵和导数。由此产生的3x3线性系统可以以最小成本求解。如果偏移量$\hat{\mathbf{x}}$在任何维度上大于0.5，则意味着极值点更接近另一个采样点。在这种情况下，将更改采样点，并围绕该点执行插值计算。最终偏移量$\hat{\mathbf{x}}$会加到其采样点的位置上，从而得到极值点位置的插值估计。

极值处的函数值 {v*} 可用于排除低对比度的不稳定极值。这可以通过将方程（3）代入（2）得到，即

$$D(\hat{\mathbf{x}}) = D + \frac{1}{2}\frac{\partial D}{\partial \mathbf{x}}^T \hat{\mathbf{x}}.$$

在本文的实验中，所有值小于0.03的极值点均被舍弃（如前所述，我们假设图像像素值范围在[0,1]之间）。

图5展示了关键点选择对自然图像的影响。为避免画面过于杂乱，此处采用低分辨率（233×189像素）图像，并将关键点以矢量形式呈现，每个矢量标明了关键点的位置、尺度和方向（方向分配方法将在下文说明）。图5(a)为原始图像，在后续子图中将以降低对比度的形式作为背景显示。图5(b)展示了所有检测到的极值点共832个关键点。

and minima of the difference-of-Gaussian function, while (c) shows the 729 keypoints that remain following removal of those with a value of $|D(\hat{\mathbf{x}})|$ less than 0.03. Part (d) will be explained in the following section.

4.1 Eliminating edge responses

For stability, it is not sufficient to reject keypoints with low contrast. The difference-of-Gaussian function will have a strong response along edges, even if the location along the edge is poorly determined and therefore unstable to small amounts of noise.

A poorly defined peak in the difference-of-Gaussian function will have a large principal curvature across the edge but a small one in the perpendicular direction. The principal curvatures can be computed from a 2x2 Hessian matrix, **H**, computed at the location and scale of the keypoint:

$$\mathbf{H} = \begin{bmatrix} D_{xx} & D_{xy} \\ D_{xy} & D_{yy} \end{bmatrix} \tag{4}$$

The derivatives are estimated by taking differences of neighboring sample points.

The eigenvalues of **H** are proportional to the principal curvatures of D. Borrowing from the approach used by Harris and Stephens (1988), we can avoid explicitly computing the eigenvalues, as we are only concerned with their ratio. Let α be the eigenvalue with the largest magnitude and β be the smaller one. Then, we can compute the sum of the eigenvalues from the trace of **H** and their product from the determinant:

$$\mathrm{Tr}(\mathbf{H}) = D_{xx} + D_{yy} = \alpha + \beta,$$
$$\mathrm{Det}(\mathbf{H}) = D_{xx}D_{yy} - (D_{xy})^2 = \alpha\beta.$$

In the unlikely event that the determinant is negative, the curvatures have different signs so the point is discarded as not being an extremum. Let r be the ratio between the largest magnitude eigenvalue and the smaller one, so that $\alpha = r\beta$. Then,

$$\frac{\mathrm{Tr}(\mathbf{H})^2}{\mathrm{Det}(\mathbf{H})} = \frac{(\alpha+\beta)^2}{\alpha\beta} = \frac{(r\beta+\beta)^2}{r\beta^2} = \frac{(r+1)^2}{r},$$

which depends only on the ratio of the eigenvalues rather than their individual values. The quantity $(r+1)^2/r$ is at a minimum when the two eigenvalues are equal and it increases with r. Therefore, to check that the ratio of principal curvatures is below some threshold, r, we only need to check

$$\frac{\mathrm{Tr}(\mathbf{H})^2}{\mathrm{Det}(\mathbf{H})} < \frac{(r+1)^2}{r}.$$

This is very efficient to compute, with less than 20 floating point operations required to test each keypoint. The experiments in this paper use a value of $r = 10$, which eliminates keypoints that have a ratio between the principal curvatures greater than 10. The transition from Figure 5 (c) to (d) shows the effects of this operation.

而高斯差分函数的极小值，同时(c)部分展示了在移除那些值小于0.03的$|D(\hat{\mathbf{x}})|$后保留的729个关键点。第(d)部分将在下一节中解释。

4.1 消除边缘响应

为了稳定性，仅拒绝低对比度的关键点是不够的。高斯差分函数在边缘处会产生强烈响应，即使边缘上的位置难以精确定位，因而对少量噪声也不稳定。

高斯差分函数中定义不佳的峰值在边缘方向上会有较大的主曲率，而在垂直方向上则较小。主曲率可通过在关键点位置和尺度上计算的2x2 Hessian矩阵$\mathbf{H}$得出：

$$\mathbf{H} = \begin{bmatrix} D_{xx} & D_{xy} \\ D_{xy} & D_{yy} \end{bmatrix} \tag{4}$$

导数通过取相邻样本点的差值来估计。

H的特征值与D的主曲率成正比。借鉴Harris和Stephens（1988）使用的方法，我们可以避免显式计算特征值，因为我们只关心它们的比例。令α为幅值最大的特征值，β为较小的特征值。那么，我们可以通过H的迹计算特征值之和，并通过其行列式计算特征值之积:

$$\mathrm{Tr}(\mathbf{H}) = D_{xx} + D_{yy} = \alpha + \beta,$$
$$\mathrm{Det}(\mathbf{H}) = D_{xx}D_{yy} - (D_{xy})^2 = \alpha\beta.$$

如果行列式为负（这种情况不太可能发生），则曲率符号不同，因此该点被舍弃，不作为极值点。设 r 为最大特征值的绝对值与较小特征值的绝对值之比，使得 $\alpha = r\beta$ 。那么，

$$\frac{\mathrm{Tr}(\mathbf{H})^2}{\mathrm{Det}(\mathbf{H})} = \frac{(\alpha+\beta)^2}{\alpha\beta} = \frac{(r\beta+\beta)^2}{r\beta^2} = \frac{(r+1)^2}{r},$$

它仅依赖于特征值的比例，而非各自的数值。当两个特征值相等时，量 $(r+1)^2/r$ 达到最小值，并随 r 增大而增加。因此，要验证主曲率之比是否低于某个阈值 r，我们只需检查

$$\frac{\mathrm{Tr}(\mathbf{H})^2}{\mathrm{Det}(\mathbf{H})} < \frac{(r+1)^2}{r}.$$

计算这一步骤非常高效，测试每个关键点所需的浮点运算次数少于20次。本文实验采用$r = 10$作为阈值，该设定会剔除主曲率比值大于10的关键点。从图5(c)到(d)的转换过程展示了此操作的效果。

5 Orientation assignment

By assigning a consistent orientation to each keypoint based on local image properties, the keypoint descriptor can be represented relative to this orientation and therefore achieve invariance to image rotation. This approach contrasts with the orientation invariant descriptors of Schmid and Mohr (1997), in which each image property is based on a rotationally invariant measure. The disadvantage of that approach is that it limits the descriptors that can be used and discards image information by not requiring all measures to be based on a consistent rotation.

Following experimentation with a number of approaches to assigning a local orientation, the following approach was found to give the most stable results. The scale of the keypoint is used to select the Gaussian smoothed image, L, with the closest scale, so that all computations are performed in a scale-invariant manner. For each image sample, $L(x, y)$, at this scale, the gradient magnitude, $m(x, y)$, and orientation, $\theta(x, y)$, is precomputed using pixel differences:

$$m(x, y) = \sqrt{(L(x+1, y) - L(x-1, y))^2 + (L(x, y+1) - L(x, y-1))^2}$$

$$\theta(x, y) = \tan^{-1}((L(x, y+1) - L(x, y-1))/(L(x+1, y) - L(x-1, y)))$$

An orientation histogram is formed from the gradient orientations of sample points within a region around the keypoint. The orientation histogram has 36 bins covering the 360 degree range of orientations. Each sample added to the histogram is weighted by its gradient magnitude and by a Gaussian-weighted circular window with a σ that is 1.5 times that of the scale of the keypoint.

Peaks in the orientation histogram correspond to dominant directions of local gradients. The highest peak in the histogram is detected, and then any other local peak that is within 80% of the highest peak is used to also create a keypoint with that orientation. Therefore, for locations with multiple peaks of similar magnitude, there will be multiple keypoints created at the same location and scale but different orientations. Only about 15% of points are assigned multiple orientations, but these contribute significantly to the stability of matching. Finally, a parabola is fit to the 3 histogram values closest to each peak to interpolate the peak position for better accuracy.

Figure 6 shows the experimental stability of location, scale, and orientation assignment under differing amounts of image noise. As before the images are rotated and scaled by random amounts. The top line shows the stability of keypoint location and scale assignment. The second line shows the stability of matching when the orientation assignment is also required to be within 15 degrees. As shown by the gap between the top two lines, the orientation assignment remains accurate 95% of the time even after addition of $\pm 10\%$ pixel noise (equivalent to a camera providing less than 3 bits of precision). The measured variance of orientation for the correct matches is about 2.5 degrees, rising to 3.9 degrees for 10% noise. The bottom line in Figure 6 shows the final accuracy of correctly matching a keypoint descriptor to a database of 40,000 keypoints (to be discussed below). As this graph shows, the SIFT features are resistant to even large amounts of pixel noise, and the major cause of error is the initial location and scale detection.

5 方向分配

通过基于局部图像属性为每个关键点分配一致的方向，关键点描述符可以相对于此方向表示，从而实现图像旋转的不变性。这种方法与Schmid和Mohr（1997）的方向不变描述符形成对比，后者中每个图像属性都基于旋转不变的度量。该方法的缺点在于，它限制了可使用的描述符类型，并且由于不要求所有度量基于一致的旋转，从而丢弃了图像信息。

在尝试了多种分配局部方向的方法后，发现以下方法能产生最稳定的结果。关键点的尺度被用于选择高斯平滑图像L，该图像具有最接近的尺度，从而所有计算都以尺度不变的方式进行。对于该尺度下的每个图像样本$L(x, y)$，使用像素差分预先计算梯度幅值$m(x, y)$和方向$\theta(x, y)$：

$$m(x, y) = \sqrt{(L(x+1, y) - L(x-1, y))^2 + (L(x, y+1) - L(x, y-1))^2}$$

$$\theta(x, y) = \tan^{-1}((L(x, y+1) - L(x, y-1))/(L(x+1, y) - L(x-1, y)))$$

方向直方图由关键点周围区域内样本点的梯度方向构成。该方向直方图包含36个柱，覆盖360度的方向范围。添加到直方图中的每个样本都按其梯度幅值以及一个高斯加权圆形窗口进行加权，该窗口的σ是关键点尺度的1.5倍。

方向直方图中的峰值对应局部梯度的主导方向。检测直方图中的最高峰，然后使用任何其他位于最高峰80%范围内的局部峰值，以该方向创建另一个关键点。因此，对于具有多个相似幅值峰值的位置，将在相同位置和尺度但不同方向上创建多个关键点。大约只有15%的点被分配了多个方向，但这些点对匹配的稳定性有显著贡献。最后，对每个峰值最接近的3个直方图值拟合抛物线，以插值峰值位置，从而提高精度。

图6展示了在不同程度的图像噪声下，位置、尺度和方向分配的实验稳定性。与之前一样，图像被随机旋转和缩放。顶线显示了关键点位置和尺度分配的稳定性。第二线显示了当方向分配也要求在15度以内时的匹配稳定性。如顶两条线之间的间隙所示，即使在添加了±10%像素噪声（相当于相机提供少于3位精度）后，方向分配仍保持95%的准确率。正确匹配的方向测量方差约为2.5度，在10%噪声下升至3.9度。图6的底线显示了将关键点描述符正确匹配到包含40,000个关键点的数据库的最终准确率（将在下文讨论）。如图所示，SIFT特征即使在大幅像素噪声下也表现出较强的鲁棒性，而误差的主要来源是初始位置和尺度检测。

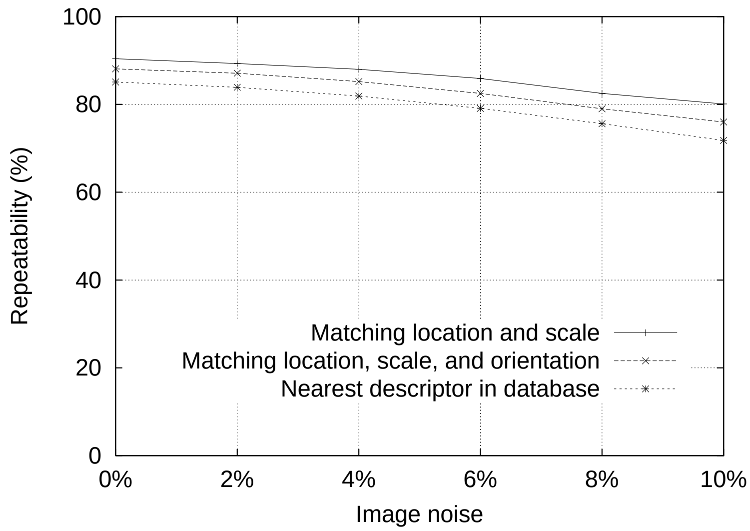

Figure 6: The top line in the graph shows the percent of keypoint locations and scales that are repeatably detected as a function of pixel noise. The second line shows the repeatability after also requiring agreement in orientation. The bottom line shows the final percent of descriptors correctly matched to a large database.

6 The local image descriptor

The previous operations have assigned an image location, scale, and orientation to each keypoint. These parameters impose a repeatable local 2D coordinate system in which to describe the local image region, and therefore provide invariance to these parameters. The next step is to compute a descriptor for the local image region that is highly distinctive yet is as invariant as possible to remaining variations, such as change in illumination or 3D viewpoint.

One obvious approach would be to sample the local image intensities around the keypoint at the appropriate scale, and to match these using a normalized correlation measure. However, simple correlation of image patches is highly sensitive to changes that cause misregistration of samples, such as affine or 3D viewpoint change or non-rigid deformations. A better approach has been demonstrated by Edelman, Intrator, and Poggio (1997). Their proposed representation was based upon a model of biological vision, in particular of complex neurons in primary visual cortex. These complex neurons respond to a gradient at a particular orientation and spatial frequency, but the location of the gradient on the retina is allowed to shift over a small receptive field rather than being precisely localized. Edelman *et al.* hypothesized that the function of these complex neurons was to allow for matching and recognition of 3D objects from a range of viewpoints. They have performed detailed experiments using 3D computer models of object and animal shapes which show that matching gradients while allowing for shifts in their position results in much better classification under 3D rotation. For example, recognition accuracy for 3D objects rotated in depth by 20 degrees increased from 35% for correlation of gradients to 94% using the complex cell model. Our implementation described below was inspired by this idea, but allows for positional shift using a different computational mechanism.

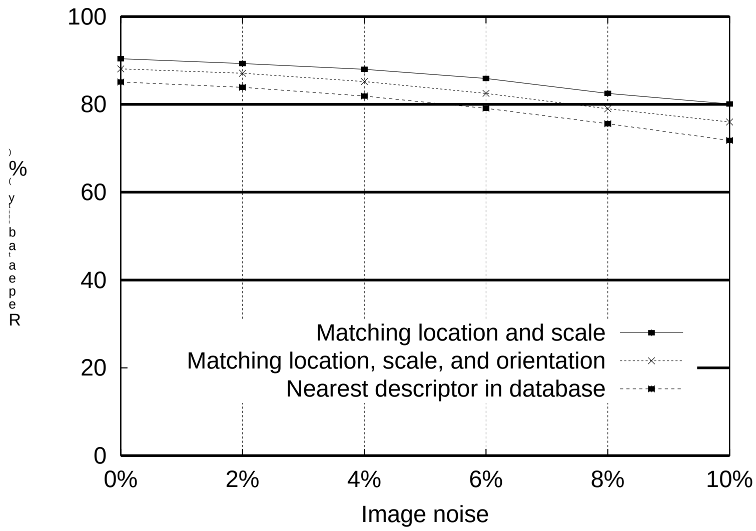

图6：图中顶线显示了关键点位置和尺度可重复检测的百分比与像素噪声的函数关系。第二条线显示了在同时要求方向一致后的可重复性。底线显示了最终正确匹配到大型数据库的描述符百分比。

6 局部图像描述符

之前的操作已经为每个关键点分配了图像位置、尺度和方向。这些参数建立了一个可重复的局部二维坐标系，用于描述局部图像区域，从而提供了对这些参数的不变性。下一步是计算局部图像区域的描述符，该描述符需具有高度区分性，同时尽可能对剩余的变化（如光照变化或三维视角变化）保持不变。

一种显而易见的方法是，在适当的尺度上对关键点周围的局部图像强度进行采样，并使用归一化相关度量进行匹配。然而，简单的图像块相关对于导致样本错配的变化（如仿射或三维视角变化或非刚性变形）极为敏感。Edelman、Intrator和Poggio（1997）展示了一种更好的方法。他们提出的表示基于生物视觉模型，特别是初级视觉皮层中的复杂神经元。这些复杂神经元对特定方向和空间频率的梯度作出响应，但梯度在视网膜上的位置允许在一个小的感受野内移动，而不是精确定位。Edelman *et al.* 假设这些复杂神经元的功能是允许从不同视角匹配和识别三维物体。他们使用物体和动物形状的三维计算机模型进行了详细实验，结果表明，在允许梯度位置移动的情况下进行匹配，能在三维旋转下实现更好的分类。例如，对于深度旋转20度的三维物体，识别准确率从梯度相关的35%提升到使用复杂细胞模型后的94%。我们下文描述的实现方法受此启发，但通过不同的计算机制允许位置移动。

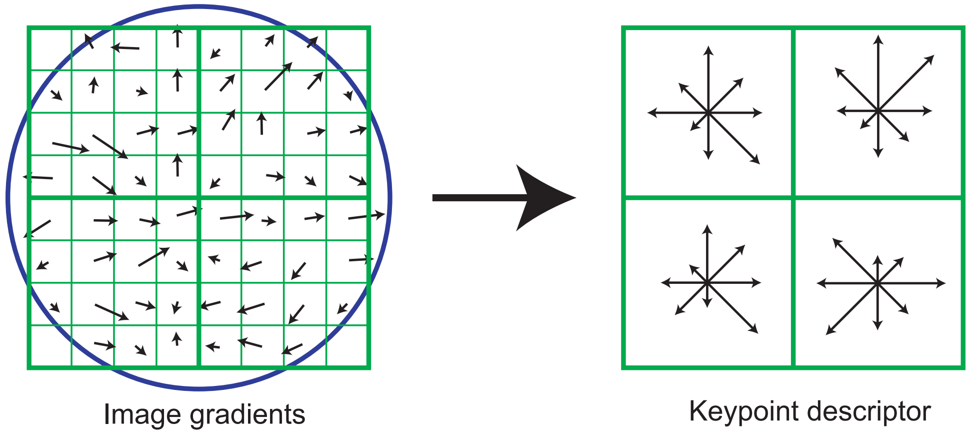

Figure 7: A keypoint descriptor is created by first computing the gradient magnitude and orientation at each image sample point in a region around the keypoint location, as shown on the left. These are weighted by a Gaussian window, indicated by the overlaid circle. These samples are then accumulated into orientation histograms summarizing the contents over 4x4 subregions, as shown on the right, with the length of each arrow corresponding to the sum of the gradient magnitudes near that direction within the region. This figure shows a 2x2 descriptor array computed from an 8x8 set of samples, whereas the experiments in this paper use 4x4 descriptors computed from a 16x16 sample array.

6.1 Descriptor representation

Figure 7 illustrates the computation of the keypoint descriptor. First the image gradient magnitudes and orientations are sampled around the keypoint location, using the scale of the keypoint to select the level of Gaussian blur for the image. In order to achieve orientation invariance, the coordinates of the descriptor and the gradient orientations are rotated relative to the keypoint orientation. For efficiency, the gradients are precomputed for all levels of the pyramid as described in Section 5. These are illustrated with small arrows at each sample location on the left side of Figure 7.

A Gaussian weighting function with σ equal to one half the width of the descriptor window is used to assign a weight to the magnitude of each sample point. This is illustrated with a circular window on the left side of Figure 7, although, of course, the weight falls off smoothly. The purpose of this Gaussian window is to avoid sudden changes in the descriptor with small changes in the position of the window, and to give less emphasis to gradients that are far from the center of the descriptor, as these are most affected by misregistration errors.

The keypoint descriptor is shown on the right side of Figure 7. It allows for significant shift in gradient positions by creating orientation histograms over 4x4 sample regions. The figure shows eight directions for each orientation histogram, with the length of each arrow corresponding to the magnitude of that histogram entry. A gradient sample on the left can shift up to 4 sample positions while still contributing to the same histogram on the right, thereby achieving the objective of allowing for larger local positional shifts.

It is important to avoid all boundary affects in which the descriptor abruptly changes as a sample shifts smoothly from being within one histogram to another or from one orientation to another. Therefore, trilinear interpolation is used to distribute the value of each gradient sample into adjacent histogram bins. In other words, each entry into a bin is multiplied by a weight of $1 - d$ for each dimension, where d is the distance of the sample from the central value of the bin as measured in units of the histogram bin spacing.

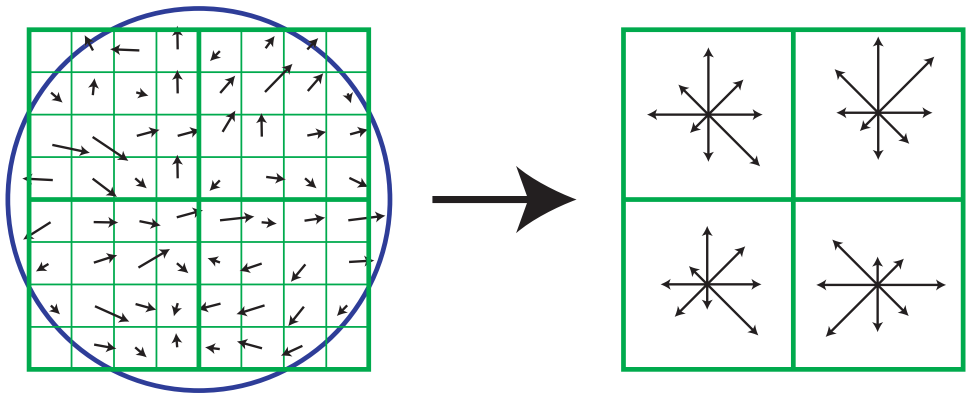

图7：关键点描述符的生成首先通过在关键点位置周围的区域内计算每个图像采样点的梯度幅值和方向，如左图所示。这些值通过一个高斯窗口进行加权，如叠加的圆所示。随后，这些采样点被累积到方向直方图中，汇总了4x4子区域内的内容，如右图所示，每个箭头的长度对应于该区域内该方向附近梯度幅值的总和。此图展示了一个由8x8采样集计算得到的2x2描述符阵列，而本文实验使用的是由16x16采样阵列计算得到的4x4描述符。

6.1 描述符表示

图7展示了关键点描述符的计算过程。首先，在关键点位置周围采样图像梯度幅值和方向，并利用关键点的尺度选择图像的高斯模糊级别。为实现方向不变性，描述符的坐标和梯度方向会相对于关键点方向进行旋转。为提高效率，如第5节所述，梯度已在金字塔的所有层级上预先计算完成。这些梯度在图7左侧的每个采样点上用小箭头示意表示。

采用一个高斯加权函数，其σ等于描述符窗口宽度的一半，用于为每个采样点的幅值分配权重。图7左侧的圆形窗口展示了这一过程，尽管权重实际上是平滑衰减的。该高斯窗口的目的是避免描述符因窗口位置的微小变化而发生突变，并降低远离描述符中心的梯度权重，因为这些梯度最容易受到配准误差的影响。

关键点描述符如图7右侧所示。它通过在4x4采样区域上创建方向直方图，允许梯度位置发生显著偏移。图中每个方向直方图显示八个方向，箭头的长度对应直方图条目的幅值。左侧的梯度样本最多可偏移4个采样位置，同时仍对右侧同一直方图作出贡献，从而实现允许更大局部位置偏移的目标。

避免所有边界效应至关重要，即当样本从一个直方图平滑过渡到另一个，或从一个方向平滑过渡到另一个方向时，描述符不应发生突变。因此，采用三线性插值将每个梯度样本的值分配到相邻的直方图区间中。换言之，每个区间条目的计入值会乘以一个权重 $1-d$（针对每个维度），其中d表示样本与该区间中心值之间的距离，以直方图区间间距为单位度量。

The descriptor is formed from a vector containing the values of all the orientation histogram entries, corresponding to the lengths of the arrows on the right side of Figure 7. The figure shows a 2x2 array of orientation histograms, whereas our experiments below show that the best results are achieved with a 4x4 array of histograms with 8 orientation bins in each. Therefore, the experiments in this paper use a 4x4x8 = 128 element feature vector for each keypoint.

Finally, the feature vector is modified to reduce the effects of illumination change. First, the vector is normalized to unit length. A change in image contrast in which each pixel value is multiplied by a constant will multiply gradients by the same constant, so this contrast change will be canceled by vector normalization. A brightness change in which a constant is added to each image pixel will not affect the gradient values, as they are computed from pixel differences. Therefore, the descriptor is invariant to affine changes in illumination. However, non-linear illumination changes can also occur due to camera saturation or due to illumination changes that affect 3D surfaces with differing orientations by different amounts. These effects can cause a large change in relative magnitudes for some gradients, but are less likely to affect the gradient orientations. Therefore, we reduce the influence of large gradient magnitudes by thresholding the values in the unit feature vector to each be no larger than 0.2, and then renormalizing to unit length. This means that matching the magnitudes for large gradients is no longer as important, and that the distribution of orientations has greater emphasis. The value of 0.2 was determined experimentally using images containing differing illuminations for the same 3D objects.

6.2 Descriptor testing

There are two parameters that can be used to vary the complexity of the descriptor: the number of orientations, r, in the histograms, and the width, n, of the $n \times n$ array of orientation histograms. The size of the resulting descriptor vector is rn^2. As the complexity of the descriptor grows, it will be able to discriminate better in a large database, but it will also be more sensitive to shape distortions and occlusion.

Figure 8 shows experimental results in which the number of orientations and size of the descriptor were varied. The graph was generated for a viewpoint transformation in which a planar surface is tilted by 50 degrees away from the viewer and 4% image noise is added. This is near the limits of reliable matching, as it is in these more difficult cases that descriptor performance is most important. The results show the percent of keypoints that find a correct match to the single closest neighbor among a database of 40,000 keypoints. The graph shows that a single orientation histogram ($n = 1$) is very poor at discriminating, but the results continue to improve up to a 4x4 array of histograms with 8 orientations. After that, adding more orientations or a larger descriptor can actually hurt matching by making the descriptor more sensitive to distortion. These results were broadly similar for other degrees of viewpoint change and noise, although in some simpler cases discrimination continued to improve (from already high levels) with 5x5 and higher descriptor sizes. Throughout this paper we use a 4x4 descriptor with 8 orientations, resulting in feature vectors with 128 dimensions. While the dimensionality of the descriptor may seem high, we have found that it consistently performs better than lower-dimensional descriptors on a range of matching tasks and that the computational cost of matching remains low when using the approximate nearest-neighbor methods described below.

描述符由一个包含所有方向直方图条目值的向量构成，这些值对应图7右侧箭头的长度。图中展示的是2x2的方向直方图阵列，而我们后续实验表明，最佳结果是通过4x4的直方图阵列（每个直方图包含8个方向区间）实现的。因此，本文实验为每个关键点使用4x4x8 = 128维特征向量。

最后，对特征向量进行修改以减少光照变化的影响。首先，将向量归一化为单位长度。图像对比度的变化（即每个像素值乘以一个常数）会使梯度乘以相同的常数，因此这种对比度变化将通过向量归一化被抵消。亮度变化（即在每个图像像素上添加一个常数）不会影响梯度值，因为梯度值是通过像素差异计算得出的。因此，该描述符对光照的仿射变化具有不变性。然而，非线性光照变化也可能发生，这可能是由于相机饱和，或是由于光照变化对不同方向的3D表面产生不同程度的影响。这些效应可能导致某些梯度的相对幅度发生较大变化，但不太可能影响梯度的方向。因此，我们通过将单位特征向量中的值阈值化（每个值不超过0.2），然后重新归一化为单位长度，来减小大幅值梯度的影响。这意味着大幅值梯度的匹配不再那么重要，而方向的分布则被赋予更大的权重。0.2这个值是通过对包含相同3D物体在不同光照下的图像进行实验确定的。

6.2 描述符测试

描述符的复杂度可以通过两个参数进行调整：直方图中的方向数量r，以及方向直方图$n \times n$数组的宽度n。生成的描述符向量大小为rn^2。随着描述符复杂度的增加，它能在大型数据库中实现更精准的区分，但同时也会对形状扭曲和遮挡更为敏感。

图8展示了改变方向数量和描述符大小的实验结果。该图表针对视角变换生成，其中平面表面相对于观察者倾斜50度，并添加了4%的图像噪声。这接近可靠匹配的极限，因为正是在这些更困难的情况下，描述符的性能最为关键。结果显示的是在包含40,000个关键点的数据库中，找到与最近邻正确匹配的关键点百分比。图表表明，单一方向直方图（$n = 1$）的区分能力非常差，但结果持续改善直至使用8个方向的4x4直方图阵列。此后，增加更多方向或使用更大的描述符实际上可能损害匹配效果，因为描述符对失真更加敏感。这些结果在其他程度的视角变化和噪声条件下大致相似，尽管在某些较简单的情况下，使用5x5及更大尺寸的描述符（在已经较高的水平上）区分能力持续提升。在本文中，我们使用具有8个方向的4x4描述符，从而生成128维的特征向量。虽然描述符的维度可能显得较高，但我们发现，在一系列匹配任务中，其性能始终优于低维描述符，并且在使用下文所述的近似最近邻方法时，匹配的计算成本仍然很低。

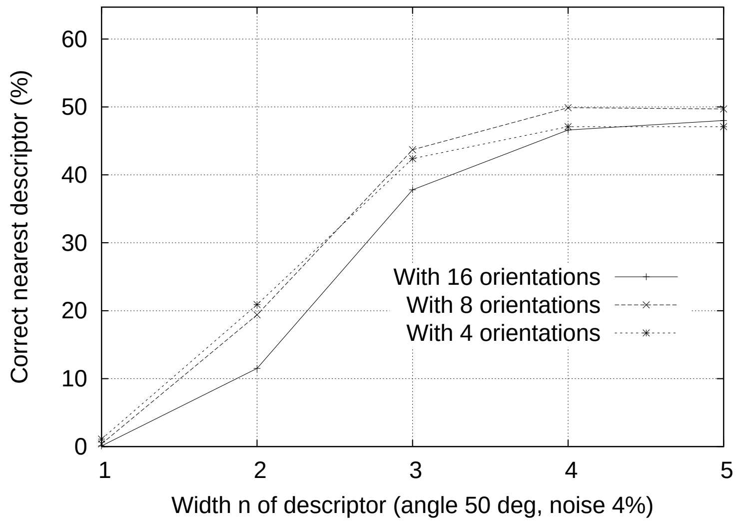

Figure 8: This graph shows the percent of keypoints giving the correct match to a database of 40,000 keypoints as a function of width of the $n \times n$ keypoint descriptor and the number of orientations in each histogram. The graph is computed for images with affine viewpoint change of 50 degrees and addition of 4% noise.

6.3 Sensitivity to affine change

The sensitivity of the descriptor to affine change is examined in Figure 9. The graph shows the reliability of keypoint location and scale selection, orientation assignment, and nearest-neighbor matching to a database as a function of rotation in depth of a plane away from a viewer. It can be seen that each stage of computation has reduced repeatability with increasing affine distortion, but that the final matching accuracy remains above 50% out to a 50 degree change in viewpoint.

To achieve reliable matching over a wider viewpoint angle, one of the affine-invariant detectors could be used to select and resample image regions, as discussed in Section 2. As mentioned there, none of these approaches is truly affine-invariant, as they all start from initial feature locations determined in a non-affine-invariant manner. In what appears to be the most affine-invariant method, Mikolajczyk (2002) has proposed and run detailed experiments with the Harris-affine detector. He found that its keypoint repeatability is below that given here out to about a 50 degree viewpoint angle, but that it then retains close to 40% repeatability out to an angle of 70 degrees, which provides better performance for extreme affine changes. The disadvantages are a much higher computational cost, a reduction in the number of keypoints, and poorer stability for small affine changes due to errors in assigning a consistent affine frame under noise. In practice, the allowable range of rotation for 3D objects is considerably less than for planar surfaces, so affine invariance is usually not the limiting factor in the ability to match across viewpoint change. If a wide range of affine invariance is desired, such as for a surface that is known to be planar, then a simple solution is to adopt the approach of Pritchard and Heidrich (2003) in which additional SIFT features are generated from 4 affine-transformed versions of the training image corresponding to 60 degree viewpoint changes. This allows for the use of standard SIFT features with no additional cost when processing the image to be recognized, but results in an increase in the size of the feature database by a factor of 3.

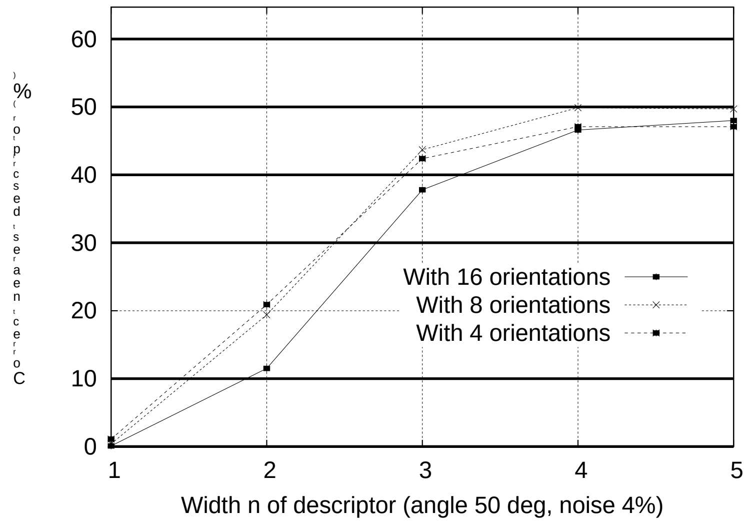

图8：该图表展示了关键点与包含40,000个关键点的数据库正确匹配的百分比，该百分比随 $n \times n$关键点描述符的宽度以及每个直方图中方向数量的变化而变化。此图表基于图像在50度仿射视角变化并添加4%噪声的条件下计算得出。

6.3 对仿射变换的敏感性

描述符对仿射变化的敏感性在图9中进行了检验。该图显示了关键点位置与尺度选择、方向分配以及与数据库的最近邻匹配的可靠性，这些均随平面相对于观察者的深度旋转而变化。可以看出，随着仿射变形的增加，计算的每个阶段的可重复性都有所降低，但即使在视角变化达到50度时，最终匹配的准确率仍保持在50%以上。

为了在更广的视角范围内实现可靠的匹配，可以采用一种仿射不变性检测器来选择和重采样图像区域，如第2节所述。如前文所提，这些方法均非真正的仿射不变方法，因为它们都始于以非仿射不变方式确定的初始特征位置。在目前最具仿射不变性的方法中，Mikolajczyk（2002）提出并利用Harris-仿射检测器进行了详细实验。他发现，在视角角度约50度以内时，其关键点重复性低于本文方法，但在视角达到70度时仍能保持接近40%的重复性，这为极端仿射变换提供了更好的性能。其缺点在于计算成本显著更高、关键点数量减少，以及因噪声下难以分配一致的仿射框架而导致对小范围仿射变化的稳定性较差。实际上，三维物体允许的旋转范围远小于平面表面，因此仿射不变性通常不是跨视角匹配能力的主要限制因素。若需要广泛的仿射不变性（例如已知表面为平面的情况），可采用Pritchard和Heidrich（2003）的简易方案：通过生成训练图像的4个仿射变换版本（对应60度视角变化）来创建额外的SIFT特征。这样在识别待处理图像时，可使用标准SIFT特征且无需额外成本，但会导致特征数据库的规模扩大至原来的3倍。

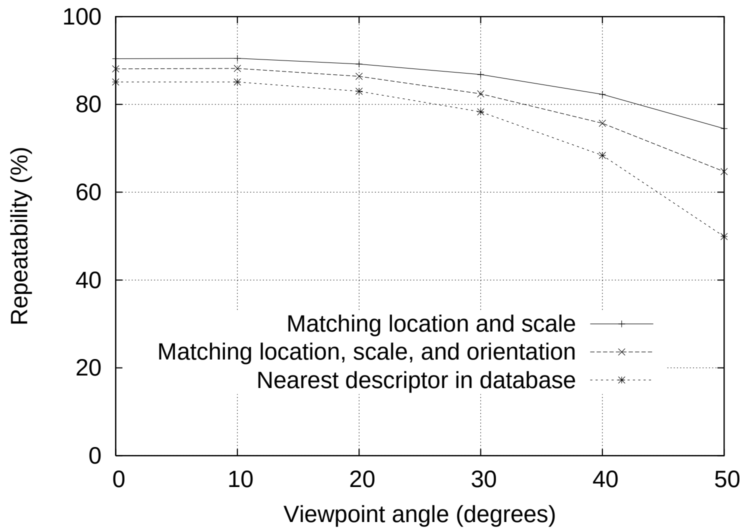

Figure 9: This graph shows the stability of detection for keypoint location, orientation, and final matching to a database as a function of affine distortion. The degree of affine distortion is expressed in terms of the equivalent viewpoint rotation in depth for a planar surface.

6.4 Matching to large databases

An important remaining issue for measuring the distinctiveness of features is how the reliability of matching varies as a function of the number of features in the database being matched. Most of the examples in this paper are generated using a database of 32 images with about 40,000 keypoints. Figure 10 shows how the matching reliability varies as a function of database size. This figure was generated using a larger database of 112 images, with a viewpoint depth rotation of 30 degrees and 2% image noise in addition to the usual random image rotation and scale change.

The dashed line shows the portion of image features for which the nearest neighbor in the database was the correct match, as a function of database size shown on a logarithmic scale. The leftmost point is matching against features from only a single image while the rightmost point is selecting matches from a database of all features from the 112 images. It can be seen that matching reliability does decrease as a function of the number of distractors, yet all indications are that many correct matches will continue to be found out to very large database sizes.

The solid line is the percentage of keypoints that were identified at the correct matching location and orientation in the transformed image, so it is only these points that have any chance of having matching descriptors in the database. The reason this line is flat is that the test was run over the full database for each value, while only varying the portion of the database used for distractors. It is of interest that the gap between the two lines is small, indicating that matching failures are due more to issues with initial feature localization and orientation assignment than to problems with feature distinctiveness, even out to large database sizes.

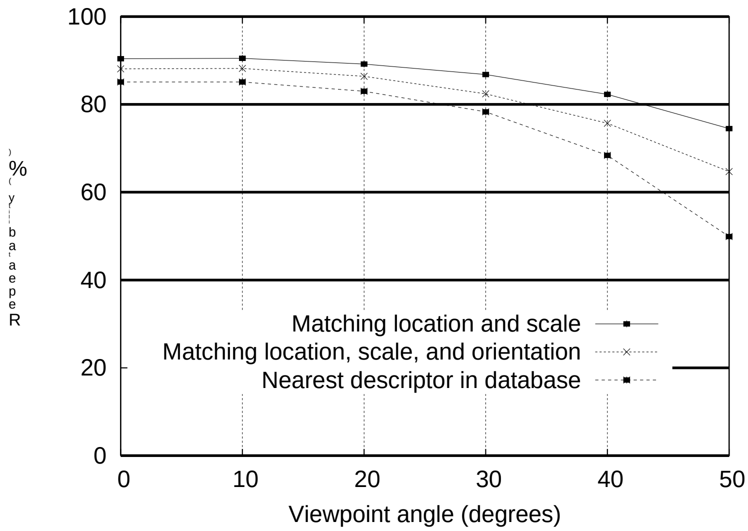

图9：该图展示了关键点位置、方向以及与数据库最终匹配的检测稳定性随仿射失真变化的情况。仿射失真程度通过平面表面在深度方向上的等效视角旋转来表示。

6.4 与大型数据库匹配

衡量特征独特性时，一个重要的遗留问题是匹配可靠性如何随数据库中待匹配特征数量的变化而变化。本文大多数示例基于包含32张图像、约4万个关键点的数据库生成。图10展示了匹配可靠性如何随数据库规模变化。该图表通过扩展数据库（112张图像）生成，除常规随机图像旋转与尺度变化外，还包含30度视角深度旋转及2%的图像噪声。

虚线显示了图像特征中最近邻匹配正确的部分，其随数据库大小的变化以对数尺度呈现。最左侧的点表示仅与单张图像的特征进行匹配，而最右侧的点表示从包含112张图像全部特征的数据库中选取匹配。可以看出，匹配可靠性确实会随着干扰项数量的增加而下降，但所有迹象表明，即使数据库规模极大，仍能持续找到大量正确匹配。

实线表示在变换后的图像中，在正确匹配位置和方向上被识别的关键点百分比，因此只有这些点才有可能在数据库中找到匹配的描述符。这条线之所以平坦，是因为测试针对每个值都在完整数据库上运行，而仅改变用于干扰项的数据库部分。值得注意的是，两条线之间的差距很小，这表明匹配失败更多是由于初始特征定位和方向分配的问题，而非特征独特性不足所致，即使在大规模数据库中也如此。

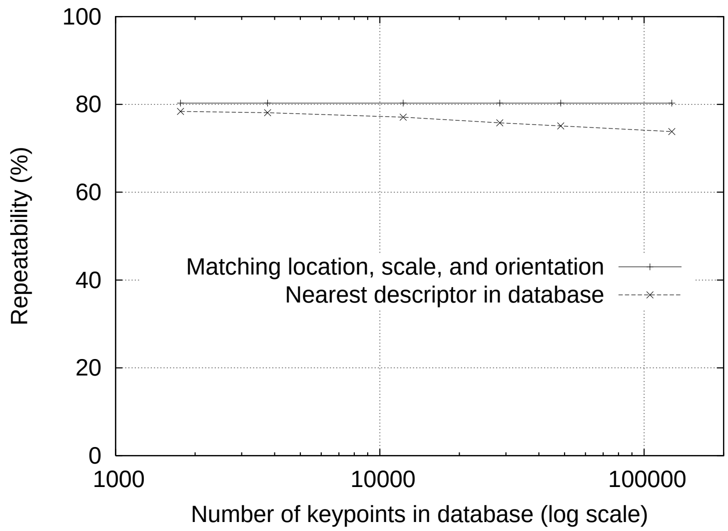

Figure 10: The dashed line shows the percent of keypoints correctly matched to a database as a function of database size (using a logarithmic scale). The solid line shows the percent of keypoints assigned the correct location, scale, and orientation. Images had random scale and rotation changes, an affine transform of 30 degrees, and image noise of 2% added prior to matching.

7 Application to object recognition

The major topic of this paper is the derivation of distinctive invariant keypoints, as described above. To demonstrate their application, we will now give a brief description of their use for object recognition in the presence of clutter and occlusion. More details on applications of these features to recognition are available in other papers (Lowe, 1999; Lowe, 2001; Se, Lowe and Little, 2002).

Object recognition is performed by first matching each keypoint independently to the database of keypoints extracted from training images. Many of these initial matches will be incorrect due to ambiguous features or features that arise from background clutter. Therefore, clusters of at least 3 features are first identified that agree on an object and its pose, as these clusters have a much higher probability of being correct than individual feature matches. Then, each cluster is checked by performing a detailed geometric fit to the model, and the result is used to accept or reject the interpretation.

7.1 Keypoint matching

The best candidate match for each keypoint is found by identifying its nearest neighbor in the database of keypoints from training images. The nearest neighbor is defined as the keypoint with minimum Euclidean distance for the invariant descriptor vector as was described in Section 6.

However, many features from an image will not have any correct match in the training database because they arise from background clutter or were not detected in the training images. Therefore, it would be useful to have a way to discard features that do not have any good match to the database. A global threshold on distance to the closest feature does not perform well, as some descriptors are much more discriminative than others. A more effective measure is obtained by comparing the distance of the closest neighbor to that of the

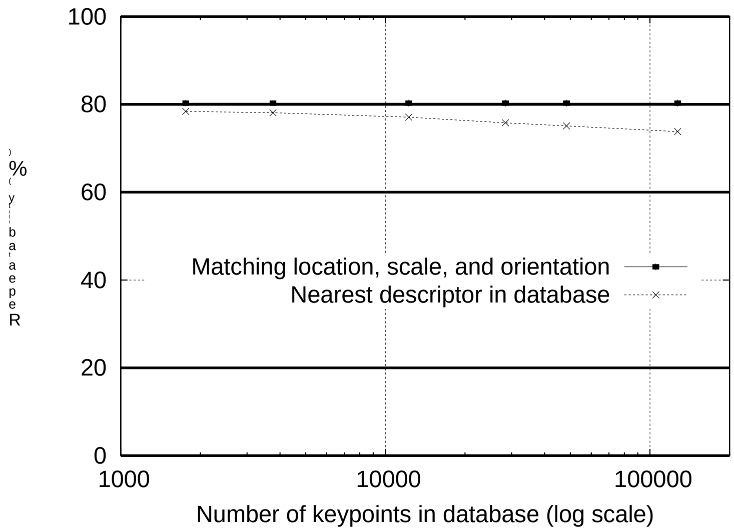

图10：虚线显示了关键点与数据库正确匹配的百分比随数据库大小（使用对数刻度）的变化关系。实线则显示了被分配正确位置、尺度和方向的关键点百分比。图像在匹配前添加了随机尺度和旋转变化、30度的仿射变换以及2%的图像噪声。

7 应用于物体识别

本文的主要议题是推导如上所述的独特不变关键点。为展示其应用，我们将简要描述其在存在杂乱和遮挡情况下用于物体识别的方法。关于这些特征在识别中应用的更多细节，可参阅其他论文（Lowe, 1999; Lowe, 2001; Se, Lowe and Little, 2002）。

物体识别首先通过将每个关键点独立地与从训练图像中提取的关键点数据库进行匹配来完成。由于模糊特征或背景杂波产生的特征，许多初始匹配将是错误的。因此，首先识别出至少3个特征组成的簇，这些簇在物体及其姿态上达成一致，因为这些簇比单个特征匹配具有更高的正确概率。接着，通过执行与模型的详细几何拟合来检查每个簇，并根据结果接受或拒绝该解释。

7.1 关键点匹配

每个关键点的最佳候选匹配是通过在训练图像的关键点数据库中识别其最近邻来找到的。最近邻被定义为具有最小欧几里得距离的不变描述符向量所对应的关键点，如第6节所述。

然而，图像中的许多特征在训练数据库中不会有任何正确匹配，因为它们来自背景杂波或在训练图像中未被检测到。因此，需要一种方法来丢弃那些在数据库中没有良好匹配的特征。对最近邻特征距离设置全局阈值效果不佳，因为某些描述符比其他描述符更具区分性。一种更有效的度量方法是通过比较最近邻的距离与{v*}的距离来获得。

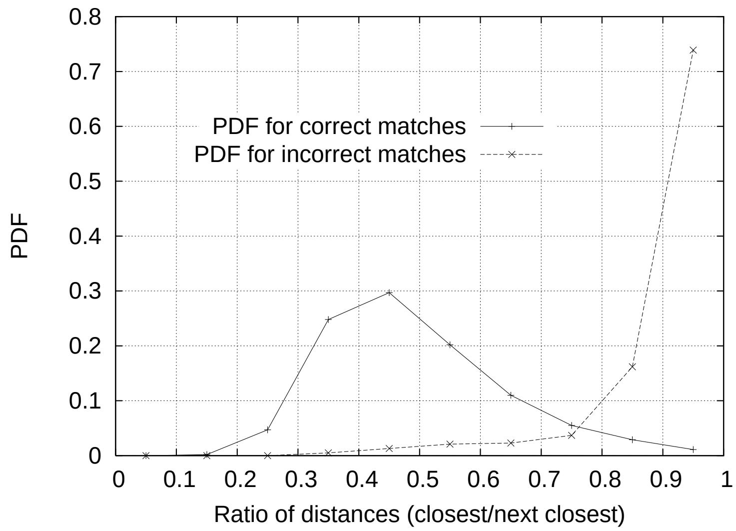

Figure 11: The probability that a match is correct can be determined by taking the ratio of distance from the closest neighbor to the distance of the second closest. Using a database of 40,000 keypoints, the solid line shows the PDF of this ratio for correct matches, while the dotted line is for matches that were incorrect.

second-closest neighbor. If there are multiple training images of the same object, then we define the second-closest neighbor as being the closest neighbor that is known to come from a different object than the first, such as by only using images known to contain different objects. This measure performs well because correct matches need to have the closest neighbor significantly closer than the closest incorrect match to achieve reliable matching. For false matches, there will likely be a number of other false matches within similar distances due to the high dimensionality of the feature space. We can think of the second-closest match as providing an estimate of the density of false matches within this portion of the feature space and at the same time identifying specific instances of feature ambiguity.

Figure 11 shows the value of this measure for real image data. The probability density functions for correct and incorrect matches are shown in terms of the ratio of closest to second-closest neighbors of each keypoint. Matches for which the nearest neighbor was a correct match have a PDF that is centered at a much lower ratio than that for incorrect matches. For our object recognition implementation, we reject all matches in which the distance ratio is greater than 0.8, which eliminates 90% of the false matches while discarding less than 5% of the correct matches. This figure was generated by matching images following random scale and orientation change, a depth rotation of 30 degrees, and addition of 2% image noise, against a database of 40,000 keypoints.

7.2 Efficient nearest neighbor indexing

No algorithms are known that can identify the exact nearest neighbors of points in high dimensional spaces that are any more efficient than exhaustive search. Our keypoint descriptor has a 128-dimensional feature vector, and the best algorithms, such as the k-d tree (Friedman *et al.,* 1977) provide no speedup over exhaustive search for more than about 10 dimensional spaces. Therefore, we have used an approximate algorithm, called the Best-Bin-First (BBF) algorithm (Beis and Lowe, 1997). This is approximate in the sense that it returns the closest

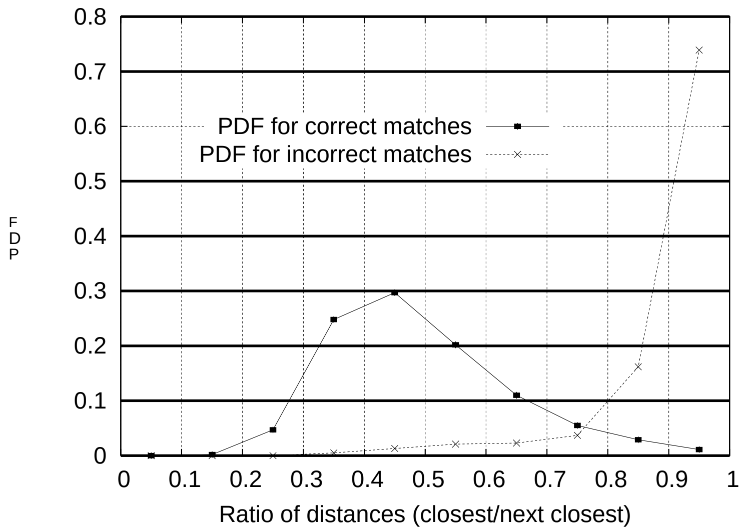

图11：匹配正确的概率可以通过计算最近邻距离与次近邻距离的比值来确定。使用包含40,000个关键点的数据库时，实线表示正确匹配的比值概率密度函数，而虚线表示错误匹配的情况。

第二近邻。如果同一物体有多个训练图像，那么我们将第二近邻定义为已知与第一近邻来自不同物体的最近邻，例如仅使用已知包含不同物体的图像。这种度量方法表现良好，因为正确匹配需要其最近邻明显比最近的不正确匹配更接近，才能实现可靠匹配。对于错误匹配，由于特征空间的高维度，很可能在相似距离内存在许多其他错误匹配。我们可以将第二近匹配视为对此部分特征空间内错误匹配密度的估计，同时识别特征模糊的具体实例。

图11展示了该度量在真实图像数据上的取值。正确匹配与错误匹配的概率密度函数以每个关键点最近邻与次近邻的比值表示。最近邻为正确匹配的对应点，其概率密度函数集中在比错误匹配低得多的比值处。在我们的物体识别实现中，我们拒绝所有距离比值大于0.8的匹配，此举可消除90%的错误匹配，同时仅舍弃不足5%的正确匹配。本图通过将图像在随机尺度与方向变化、30度深度旋转及添加2%图像噪声后，与包含40,000个关键点的数据库进行匹配而生成。

7.2 高效最近邻索引

目前尚无已知算法能够以比穷举搜索更高效的方式识别高维空间中点的确切最近邻。我们的关键点描述符具有128维特征向量，而最佳算法（如k-d树（Friedman *et al.*, 1977））在超过约10维的空间中无法提供比穷举搜索更快的速度。因此，我们采用了一种近似算法，称为最佳优先分箱（BBF）算法（Beis和Lowe，1997）。该算法的近似性在于它返回的是最接近的

neighbor with high probability.

The BBF algorithm uses a modified search ordering for the k-d tree algorithm so that bins in feature space are searched in the order of their closest distance from the query location. This priority search order was first examined by Arya and Mount (1993), and they provide further study of its computational properties in (Arya *et al.*, 1998). This search order requires the use of a heap-based priority queue for efficient determination of the search order. An approximate answer can be returned with low cost by cutting off further search after a specific number of the nearest bins have been explored. In our implementation, we cut off search after checking the first 200 nearest-neighbor candidates. For a database of 100,000 keypoints, this provides a speedup over exact nearest neighbor search by about 2 orders of magnitude yet results in less than a 5% loss in the number of correct matches. One reason the BBF algorithm works particularly well for this problem is that we only consider matches in which the nearest neighbor is less than 0.8 times the distance to the second-nearest neighbor (as described in the previous section), and therefore there is no need to exactly solve the most difficult cases in which many neighbors are at very similar distances.

7.3 Clustering with the Hough transform

To maximize the performance of object recognition for small or highly occluded objects, we wish to identify objects with the fewest possible number of feature matches. We have found that reliable recognition is possible with as few as 3 features. A typical image contains 2,000 or more features which may come from many different objects as well as background clutter. While the distance ratio test described in Section 7.1 will allow us to discard many of the false matches arising from background clutter, this does not remove matches from other valid objects, and we often still need to identify correct subsets of matches containing less than 1% inliers among 99% outliers. Many well-known robust fitting methods, such as RANSAC or Least Median of Squares, perform poorly when the percent of inliers falls much below 50%. Fortunately, much better performance can be obtained by clustering features in pose space using the Hough transform (Hough, 1962; Ballard, 1981; Grimson 1990).

The Hough transform identifies clusters of features with a consistent interpretation by using each feature to vote for all object poses that are consistent with the feature. When clusters of features are found to vote for the same pose of an object, the probability of the interpretation being correct is much higher than for any single feature. Each of our keypoints specifies 4 parameters: 2D location, scale, and orientation, and each matched keypoint in the database has a record of the keypoint's parameters relative to the training image in which it was found. Therefore, we can create a Hough transform entry predicting the model location, orientation, and scale from the match hypothesis. This prediction has large error bounds, as the similarity transform implied by these 4 parameters is only an approximation to the full 6 degree-of-freedom pose space for a 3D object and also does not account for any non-rigid deformations. Therefore, we use broad bin sizes of 30 degrees for orientation, a factor of 2 for scale, and 0.25 times the maximum projected training image dimension (using the predicted scale) for location. To avoid the problem of boundary effects in bin assignment, each keypoint match votes for the 2 closest bins in each dimension, giving a total of 16 entries for each hypothesis and further broadening the pose range.

In most implementations of the Hough transform, a multi-dimensional array is used to represent the bins. However, many of the potential bins will remain empty, and it is difficult to compute the range of possible bin values due to their mutual dependence (for example,

高概率的邻居。

BBF算法对k-d树算法采用了改进的搜索顺序，使得特征空间中的区间按照与查询位置的最小距离顺序进行搜索。这种优先级搜索顺序最早由Arya和Mount（1993年）提出，并在（Arya *et al.*, 1998年）中对其计算特性进行了进一步研究。该搜索顺序需要使用基于堆的优先级队列来高效确定搜索顺序。通过限制在探索特定数量的最近区间后停止进一步搜索，可以以较低成本返回近似答案。在我们的实现中，我们在检查前200个最近邻候选点后停止搜索。对于包含10万个关键点的数据库，这比精确最近邻搜索提高了约两个数量级的速度，同时正确匹配数量损失不到5%。BBF算法在此问题上表现优异的一个原因是，我们仅考虑最近邻距离小于次近邻距离0.8倍的匹配（如前一节所述），因此无需精确求解那些多个邻域距离极其接近的最困难情况。

7.3 使用霍夫变换进行聚类

为了最大化小型或高度遮挡物体的识别性能，我们希望以尽可能少的特征匹配数量来识别物体。我们发现，仅需3个特征即可实现可靠的识别。典型图像包含2000个或更多特征，这些特征可能来自许多不同物体以及背景杂波。虽然第7.1节中描述的距离比测试可以让我们丢弃许多由背景杂波产生的错误匹配，但这无法消除来自其他有效物体的匹配，我们通常仍需要从99%离群点中识别出包含不足1%内点的正确匹配子集。当内点比例远低于50%时，许多著名的鲁棒拟合方法（如RANSAC或最小中值平方法）表现不佳。幸运的是，通过使用霍夫变换在姿态空间中对特征进行聚类（Hough, 1962; Ballard, 1981; Grimson 1990），可以获得更好的性能。

霍夫变换通过让每个特征对所有与其一致的对象姿态进行投票，来识别具有一致解释的特征簇。当发现特征簇对同一对象姿态投票时，该解释正确的概率远高于任何单一特征。我们的每个关键点指定了4个参数：二维位置、尺度和方向，而数据库中每个匹配的关键点都记录了该关键点相对于其所在训练图像的参数。因此，我们可以根据匹配假设创建一个霍夫变换条目，预测模型的位置、方向和尺度。由于这4个参数隐含的相似变换仅是三维物体完整6自由度姿态空间的近似，且未考虑任何非刚性形变，该预测具有较大的误差范围。因此，我们使用30度的宽方向区间、2倍的尺度因子，以及0.25倍的最大投影训练图像尺寸（使用预测尺度）作为位置区间。为避免区间分配中的边界效应问题，每个关键点匹配在每个维度上为最近的2个区间投票，从而每个假设产生16个条目，进一步拓宽了姿态范围。

在大多数霍夫变换的实现中，会使用多维数组来表示区间。然而，许多潜在的区间将保持为空，并且由于它们之间的相互依赖关系（例如，

the dependency of location discretization on the selected scale). These problems can be avoided by using a pseudo-random hash function of the bin values to insert votes into a one-dimensional hash table, in which collisions are easily detected.

7.4 Solution for affine parameters

The Hough transform is used to identify all clusters with at least 3 entries in a bin. Each such cluster is then subject to a geometric verification procedure in which a least-squares solution is performed for the best affine projection parameters relating the training image to the new image.

An affine transformation correctly accounts for 3D rotation of a planar surface under orthographic projection, but the approximation can be poor for 3D rotation of non-planar objects. A more general solution would be to solve for the fundamental matrix (Luong and Faugeras, 1996; Hartley and Zisserman, 2000). However, a fundamental matrix solution requires at least 7 point matches as compared to only 3 for the affine solution and in practice requires even more matches for good stability. We would like to perform recognition with as few as 3 feature matches, so the affine solution provides a better starting point and we can account for errors in the affine approximation by allowing for large residual errors. If we imagine placing a sphere around an object, then rotation of the sphere by 30 degrees will move no point within the sphere by more than 0.25 times the projected diameter of the sphere. For the examples of typical 3D objects used in this paper, an affine solution works well given that we allow residual errors up to 0.25 times the maximum projected dimension of the object. A more general approach is given in (Brown and Lowe, 2002), in which the initial solution is based on a similarity transform, which then progresses to solution for the fundamental matrix in those cases in which a sufficient number of matches are found.

The affine transformation of a model point $[x\ y]^T$ to an image point $[u\ v]^T$ can be written as

$$\begin{bmatrix} u \\ v \end{bmatrix} = \begin{bmatrix} m_1 & m_2 \\ m_3 & m_4 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} + \begin{bmatrix} t_x \\ t_y \end{bmatrix}$$

where the model translation is $[t_x\ t_y]^T$ and the affine rotation, scale, and stretch are represented by the m_i parameters.

We wish to solve for the transformation parameters, so the equation above can be rewritten to gather the unknowns into a column vector:

$$\begin{bmatrix} x & y & 0 & 0 & 1 & 0 \\ 0 & 0 & x & y & 0 & 1 \\ & & \dots & & & \\ & & \dots & & & \end{bmatrix} \begin{bmatrix} m_1 \\ m_2 \\ m_3 \\ m_4 \\ t_x \\ t_y \end{bmatrix} = \begin{bmatrix} u \\ v \\ \vdots \end{bmatrix}$$

This equation shows a single match, but any number of further matches can be added, with each match contributing two more rows to the first and last matrix. At least 3 matches are needed to provide a solution.

We can write this linear system as

$$\mathbf{Ax} = \mathbf{b}$$

位置离散化对所选尺度的依赖性）。通过使用箱值的伪随机哈希函数将投票插入一维哈希表，可以避免这些问题，在该表中碰撞很容易被检测到。

7.4 仿射参数求解

霍夫变换用于识别每个区间内至少包含3个条目的所有聚类。随后，每个此类聚类需经过几何验证流程，通过最小二乘法求解最佳仿射投影参数，以建立训练图像与新图像之间的映射关系。

仿射变换能够正确解释正交投影下平面表面的三维旋转，但对于非平面物体的三维旋转，其近似效果可能较差。更通用的解决方案是求解基本矩阵（Luong和Faugeras，1996；Hartley和Zisserman，2000）。然而，基本矩阵求解至少需要7个匹配点，而仿射解仅需3个，且在实际中需要更多匹配点以保证良好稳定性。我们希望仅用3个特征匹配点就能进行识别，因此仿射解提供了更好的起点，并且可以通过允许较大的残差来补偿仿射近似中的误差。假设围绕物体放置一个球体，那么球体旋转30度时，球内任意点的移动距离不会超过球体投影直径的0.25倍。对于本文使用的典型三维物体示例，在允许残差达到物体最大投影尺寸0.25倍的情况下，仿射解能够取得良好效果。（Brown和Lowe，2002）提出了更通用的方法：初始解基于相似变换，当匹配点数量充足时，该方法会进一步求解基本矩阵。

模型点$[x\ y]^T$到图像点$[u\ v]^T$的仿射变换可表示为
作为

$$\begin{bmatrix} u \\ v \end{bmatrix} = \begin{bmatrix} m_1 & m_2 \\ m_3 & m_4 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} + \begin{bmatrix} t_x \\ t_y \end{bmatrix}$$

其中模型变换为$[t_x\ t_y]^T$，仿射旋转、缩放和拉伸由m_i参数表示。

我们希望求解变换参数，因此上述方程可以改写，将未知数收集到列向量中：

$$\begin{bmatrix} x & y & 0 & 0 & 1 & 0 \\ 0 & 0 & x & y & 0 & 1 \\ & & \cdots & & & \\ & & \cdots & & & \end{bmatrix} \begin{bmatrix} m_1 \\ m_2 \\ m_3 \\ m_4 \\ t_x \\ t_y \end{bmatrix} = \begin{bmatrix} u \\ v \\ \vdots \end{bmatrix}$$

该方程展示了一个单一匹配，但可以添加任意数量的进一步匹配，每个匹配都会为第一个和最后一个矩阵增加两行。至少需要3个匹配才能提供一个解。

我们可以将这个线性系统写作

$$\mathbf{Ax} = \mathbf{b}$$

Figure 12: The training images for two objects are shown on the left. These can be recognized in a cluttered image with extensive occlusion, shown in the middle. The results of recognition are shown on the right. A parallelogram is drawn around each recognized object showing the boundaries of the original training image under the affine transformation solved for during recognition. Smaller squares indicate the keypoints that were used for recognition.

The least-squares solution for the parameters $\mathbf{x}$ can be determined by solving the corresponding normal equations,

$$\mathbf{x} = [\mathbf{A}^{\mathbf{T}}\mathbf{A}]^{-1}\mathbf{A}^{\mathbf{T}}\mathbf{b},$$

which minimizes the sum of the squares of the distances from the projected model locations to the corresponding image locations. This least-squares approach could readily be extended to solving for 3D pose and internal parameters of articulated and flexible objects (Lowe, 1991).

Outliers can now be removed by checking for agreement between each image feature and the model. Given the more accurate least-squares solution, we now require each match to agree within half the error range that was used for the parameters in the Hough transform bins. If fewer than 3 points remain after discarding outliers, then the match is rejected. As outliers are discarded, the least-squares solution is re-solved with the remaining points, and the process iterated. In addition, a top-down matching phase is used to add any further matches that agree with the projected model position. These may have been missed from the Hough transform bin due to the similarity transform approximation or other errors.

The final decision to accept or reject a model hypothesis is based on a detailed probabilistic model given in a previous paper (Lowe, 2001). This method first computes the expected number of false matches to the model pose, given the projected size of the model, the number of features within the region, and the accuracy of the fit. A Bayesian analysis then gives the probability that the object is present based on the actual number of matching features found. We accept a model if the final probability for a correct interpretation is greater than 0.98. For objects that project to small regions of an image, 3 features may be sufficient for reliable recognition. For large objects covering most of a heavily textured image, the expected number of false matches is higher, and as many as 10 feature matches may be necessary.

8 Recognition examples

Figure 12 shows an example of object recognition for a cluttered and occluded image containing 3D objects. The training images of a toy train and a frog are shown on the left.

图12：左侧展示了两个物体的训练图像。这些物体可在中间所示、存在大量遮挡的杂乱图像中被识别。识别结果如右侧所示。每个被识别物体周围绘制了一个平行四边形，显示了在识别过程中求解出的仿射变换下原始训练图像的边界。较小的方块表示用于识别的关键点。

参数x的最小二乘解可以通过求解相应的正规方程来确定，

$$\mathbf{x} = [\mathbf{A}^{\mathbf{T}}\mathbf{A}]^{-1}\mathbf{A}^{\mathbf{T}}\mathbf{b},$$

它最小化了从投影模型位置到对应图像位置距离的平方和。这种最小二乘法可以轻松扩展到求解铰接和柔性物体的三维姿态及内部参数（Lowe，1991）。

现在可以通过检查每个图像特征与模型之间的一致性来剔除异常值。由于获得了更精确的最小二乘解，我们现在要求每个匹配的误差范围不超过霍夫变换箱中参数所用误差范围的一半。如果剔除异常值后剩余点数少于3个，则拒绝该匹配。在剔除异常值的同时，会使用剩余点重新求解最小二乘解，并迭代此过程。此外，采用自上而下的匹配阶段来添加与投影模型位置一致的其他匹配。这些匹配可能由于相似变换近似或其他误差而在霍夫变换箱中被遗漏。

接受或拒绝模型假设的最终决定基于先前论文（Lowe，2001）中给出的详细概率模型。该方法首先根据模型的投影尺寸、区域内特征数量以及拟合精度，计算与模型姿态匹配的预期误匹配数量。随后，通过贝叶斯分析，根据实际找到的匹配特征数量得出目标存在的概率。若正确解释的最终概率大于0.98，我们便接受该模型。对于投影到图像小区域的物体，仅需3个特征即可实现可靠识别；而对于覆盖大部分高纹理图像的大型物体，预期误匹配数量较高，可能需要多达10个特征匹配才能确保准确性。

8 识别示例

图12展示了一个包含3D物体的杂乱且被遮挡图像的物体识别示例。左侧展示了玩具火车和青蛙的训练图像。

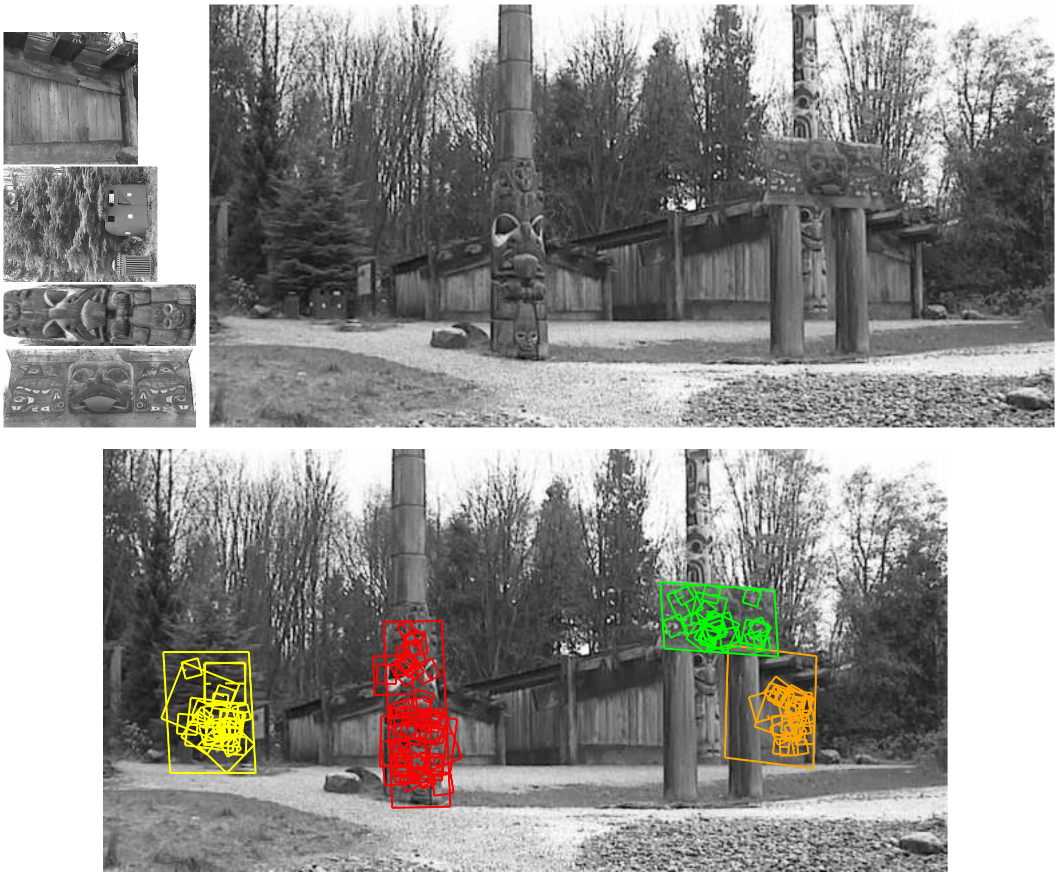

Figure 13: This example shows location recognition within a complex scene. The training images for locations are shown at the upper left and the 640x315 pixel test image taken from a different viewpoint is on the upper right. The recognized regions are shown on the lower image, with keypoints shown as squares and an outer parallelogram showing the boundaries of the training images under the affine transform used for recognition.

The middle image (of size 600x480 pixels) contains instances of these objects hidden behind others and with extensive background clutter so that detection of the objects may not be immediate even for human vision. The image on the right shows the final correct identification superimposed on a reduced contrast version of the image. The keypoints that were used for recognition are shown as squares with an extra line to indicate orientation. The sizes of the squares correspond to the image regions used to construct the descriptor. An outer parallelogram is also drawn around each instance of recognition, with its sides corresponding to the boundaries of the training images projected under the final affine transformation determined during recognition.

Another potential application of the approach is to place recognition, in which a mobile device or vehicle could identify its location by recognizing familiar locations. Figure 13 gives an example of this application, in which training images are taken of a number of locations. As shown on the upper left, these can even be of such seemingly non-distinctive items as a wooden wall or a tree with trash bins. The test image (of size 640 by 315 pixels) on the upper right was taken from a viewpoint rotated about 30 degrees around the scene from the original positions, yet the training image locations are easily recognized.

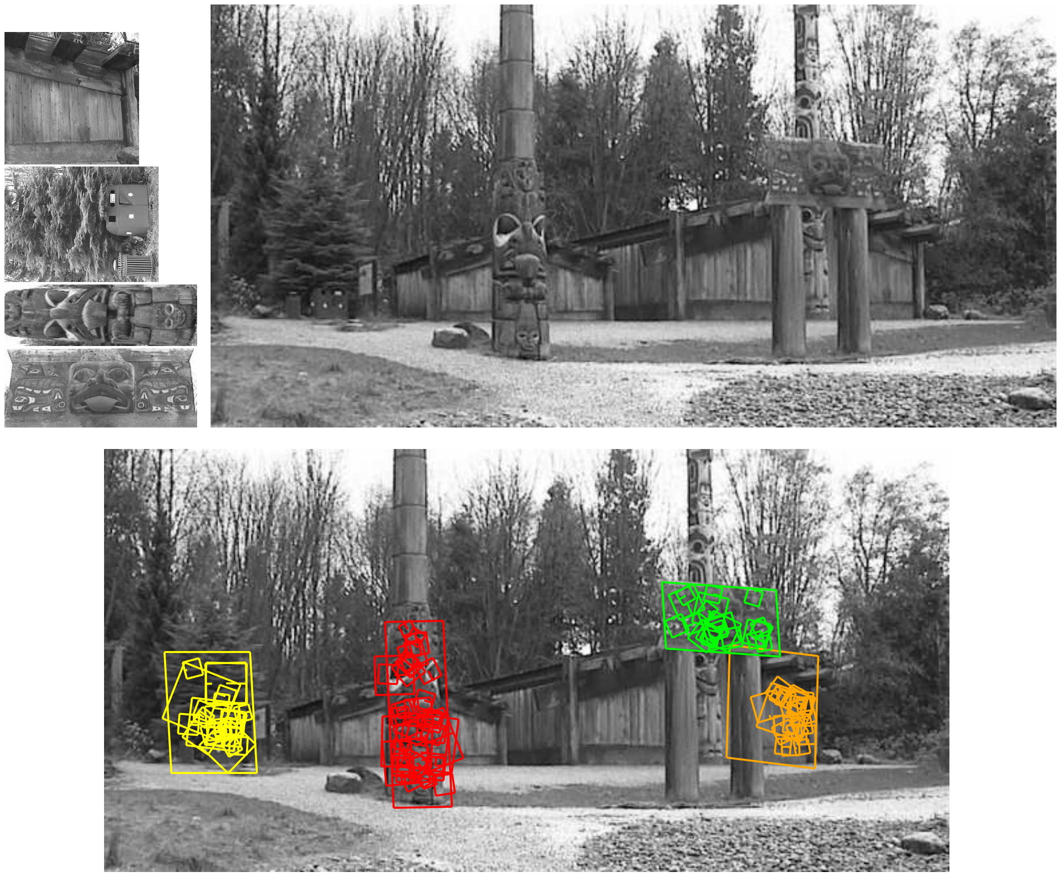

图13：此示例展示了复杂场景中的位置识别。训练位置图像显示在左上角，而从不同视角拍摄的640x315像素测试图像位于右上角。识别出的区域显示在下方的图像中，关键点以正方形标示，外部平行四边形显示了用于识别的仿射变换下训练图像的边界。

中间图像（尺寸为600x480像素）包含了这些物体的实例，它们被其他物体遮挡，并且背景杂乱无章，因此即使对于人类视觉来说，检测这些物体也可能不是立即就能完成的。右侧的图像显示了最终的正确识别结果，叠加在降低对比度的图像版本上。用于识别的关键点显示为带有额外线条指示方向的正方形。正方形的大小对应于用于构建描述符的图像区域。每个识别实例周围还绘制了一个外接平行四边形，其边对应于训练图像在识别过程中确定的最终仿射变换下投影的边界。

该方法的另一个潜在应用是地点识别，移动设备或车辆可通过识别熟悉的地点来确定自身位置。图13展示了此类应用的示例，其中训练图像采集自多个地点。如左上角所示，这些图像甚至可以拍摄看似无显著特征的物体，例如木墙或带有垃圾桶的树木。右上角的测试图像（尺寸为640×315像素）拍摄视角相对于原始位置绕场景旋转了约30度，但训练图像中的地点仍能被轻松识别。

All steps of the recognition process can be implemented efficiently, so the total time to recognize all objects in Figures 12 or 13 is less than 0.3 seconds on a 2GHz Pentium 4 processor. We have implemented these algorithms on a laptop computer with attached video camera, and have tested them extensively over a wide range of conditions. In general, textured planar surfaces can be identified reliably over a rotation in depth of up to 50 degrees in any direction and under almost any illumination conditions that provide sufficient light and do not produce excessive glare. For 3D objects, the range of rotation in depth for reliable recognition is only about 30 degrees in any direction and illumination change is more disruptive. For these reasons, 3D object recognition is best performed by integrating features from multiple views, such as with local feature view clustering (Lowe, 2001).

These keypoints have also been applied to the problem of robot localization and mapping, which has been presented in detail in other papers (Se, Lowe and Little, 2001). In this application, a trinocular stereo system is used to determine 3D estimates for keypoint locations. Keypoints are used only when they appear in all 3 images with consistent disparities, resulting in very few outliers. As the robot moves, it localizes itself using feature matches to the existing 3D map, and then incrementally adds features to the map while updating their 3D positions using a Kalman filter. This provides a robust and accurate solution to the problem of robot localization in unknown environments. This work has also addressed the problem of place recognition, in which a robot can be switched on and recognize its location anywhere within a large map (Se, Lowe and Little, 2002), which is equivalent to a 3D implementation of object recognition.

9 Conclusions

The SIFT keypoints described in this paper are particularly useful due to their distinctiveness, which enables the correct match for a keypoint to be selected from a large database of other keypoints. This distinctiveness is achieved by assembling a high-dimensional vector representing the image gradients within a local region of the image. The keypoints have been shown to be invariant to image rotation and scale and robust across a substantial range of affine distortion, addition of noise, and change in illumination. Large numbers of keypoints can be extracted from typical images, which leads to robustness in extracting small objects among clutter. The fact that keypoints are detected over a complete range of scales means that small local features are available for matching small and highly occluded objects, while large keypoints perform well for images subject to noise and blur. Their computation is efficient, so that several thousand keypoints can be extracted from a typical image with near real-time performance on standard PC hardware.

This paper has also presented methods for using the keypoints for object recognition. The approach we have described uses approximate nearest-neighbor lookup, a Hough transform for identifying clusters that agree on object pose, least-squares pose determination, and final verification. Other potential applications include view matching for 3D reconstruction, motion tracking and segmentation, robot localization, image panorama assembly, epipolar calibration, and any others that require identification of matching locations between images.

There are many directions for further research in deriving invariant and distinctive image features. Systematic testing is needed on data sets with full 3D viewpoint and illumination changes. The features described in this paper use only a monochrome intensity image, so further distinctiveness could be derived from including illumination-invariant color descriptors

识别过程的所有步骤都可以高效实现，因此在2GHz奔腾4处理器上，识别图12或图13中所有物体的总时间不到0.3秒。我们已在配备摄像头的笔记本电脑上实现了这些算法，并在各种条件下进行了广泛测试。总体而言，在任意方向深度旋转不超过50度、且光照条件能提供充足光线且不产生过度眩光的情况下，纹理平面表面可以被可靠识别。对于三维物体，可靠识别的深度旋转范围在任意方向仅约30度，且光照变化的影响更为显著。因此，三维物体识别最好通过整合多视角特征来实现，例如采用局部特征视图聚类方法（Lowe, 2001）。

这些关键点也被应用于机器人定位与地图构建问题，该问题已在其他论文中详细阐述（Se, Lowe and Little, 2001）。在此应用中，采用三目立体视觉系统来确定关键点位置的三维估计值。仅当关键点出现在所有三幅图像中且视差一致时才会被使用，从而极大减少了异常值。随着机器人移动，它通过特征匹配与现有三维地图进行自身定位，随后利用卡尔曼滤波器更新三维位置，逐步向地图中添加特征。这为机器人在未知环境中的定位问题提供了稳健而精确的解决方案。此项研究还解决了地点识别问题，即机器人可在任意位置启动并识别其在大规模地图中的方位（Se, Lowe and Little, 2002），这实质上等同于物体识别的三维实现。

9 结论

本文描述的SIFT关键点因其独特性而尤为有用，这种特性使得能够从包含大量其他关键点的数据库中准确选出匹配项。这种独特性是通过构建一个高维向量来实现的，该向量代表了图像局部区域内的梯度信息。这些关键点已被证明对图像旋转和缩放具有不变性，并且能在较大范围的仿射畸变、噪声添加以及光照变化下保持鲁棒性。从典型图像中可以提取出大量关键点，这增强了在杂乱背景中提取小目标物体的鲁棒性。关键点在完整尺度范围内被检测的特性，意味着可用于匹配小型且高度遮挡物体的局部细微特征，而大型关键点则在处理受噪声和模糊影响的图像时表现优异。其计算效率高，在标准PC硬件上能以近实时性能从典型图像中提取数千个关键点。

本文还介绍了利用关键点进行物体识别的方法。我们描述的方法采用近似最近邻查找、用于识别物体姿态一致簇的霍夫变换、最小二乘姿态确定以及最终验证。其他潜在应用包括用于三维重建的视图匹配、运动跟踪与分割、机器人定位、图像全景拼接、极线校准，以及任何需要在图像间识别匹配位置的应用。

在推导不变且独特的图像特征方面，还有许多进一步研究的方向。需要在具有完整三维视角和光照变化的数据集上进行系统测试。本文描述的特征仅使用单色强度图像，因此通过纳入光照不变的颜色描述符，可以进一步提升特征的独特性。

(Funt and Finlayson, 1995; Brown and Lowe, 2002). Similarly, local texture measures appear to play an important role in human vision and could be incorporated into feature descriptors in a more general form than the single spatial frequency used by the current descriptors. An attractive aspect of the invariant local feature approach to matching is that there is no need to select just one feature type, and the best results are likely to be obtained by using many different features, all of which can contribute useful matches and improve overall robustness.

Another direction for future research will be to individually learn features that are suited to recognizing particular objects categories. This will be particularly important for generic object classes that must cover a broad range of possible appearances. The research of Weber, Welling, and Perona (2000) and Fergus, Perona, and Zisserman (2003) has shown the potential of this approach by learning small sets of local features that are suited to recognizing generic classes of objects. In the long term, feature sets are likely to contain both prior and learned features that will be used according to the amount of training data that has been available for various object classes.

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(Funt and Finlayson, 1995; Brown and Lowe, 2002)。类似地，局部纹理度量似乎在人类视觉中扮演着重要角色，并且可以以比当前描述符所使用的单一空间频率更一般的形式融入特征描述符中。基于不变局部特征的匹配方法的一个吸引人之处在于，无需仅选择一种特征类型，而通过使用多种不同的特征——所有这些特征都能提供有效的匹配并提升整体鲁棒性——很可能获得最佳结果。

未来研究的另一个方向将是单独学习适合识别特定物体类别的特征。这对于必须涵盖广泛可能外观的通用物体类别尤为重要。Weber、Welling和Perona（2000年）以及Fergus、Perona和Zisserman（2003年）的研究通过训练适用于识别通用物体类别的小型局部特征集，展示了这种方法的潜力。从长远来看，特征集可能同时包含先验特征和习得特征，这些特征将根据针对不同物体类别可用的训练数据量来使用。

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